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S/4HANA Overview

Deloitte Technology Academy (DTA)

Agenda

Topics	Descriptions	Duration
S/4HANA Overview	S/4HANA Introduction, S/4HANA Benefits, Data Model Simplification, and S/4HANA—User Experience Simplification	8 hours
S/4HANA Architecture	SAP HANA High-Level Architecture and Features, Row Store Architecture, Column Store Architecture, Multi-Version Concurrency Control (MVCC), and Key Change Areas	
S/4HANA Migration	S/4HANA Enterprise Management, SAP S/4HANA Migration Cockpit, SAP S/4HANA Migration Object Modeler, SAP S/4HANA—Transition, SAP S/4HANA System Conversion, Maintenance Planner, Pre-Checks, Recommended Preparation for a System Conversion Project, Functional Adaptation, Performance Tuning, System Conversion, SAP S/4HANA Migration Approach, Major Phases in System Conversion to SAP S/4HANA, Recommended Migration Scenarios, Roadmap Options for S/4HANA, Greenfield vs. Brownfield Migration Analysis, and Data Consideration for SAP S/4HANA Transition	

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Learning Objectives

By the end of this session, you will be able to:

- Describe S/4HANA features
- Discuss S/4HANA benefits
- Describe SAP HANA High-Level Architecture and Features
- Describe S/4HANA migration process
- Discuss S/4HANA Roadmap Options

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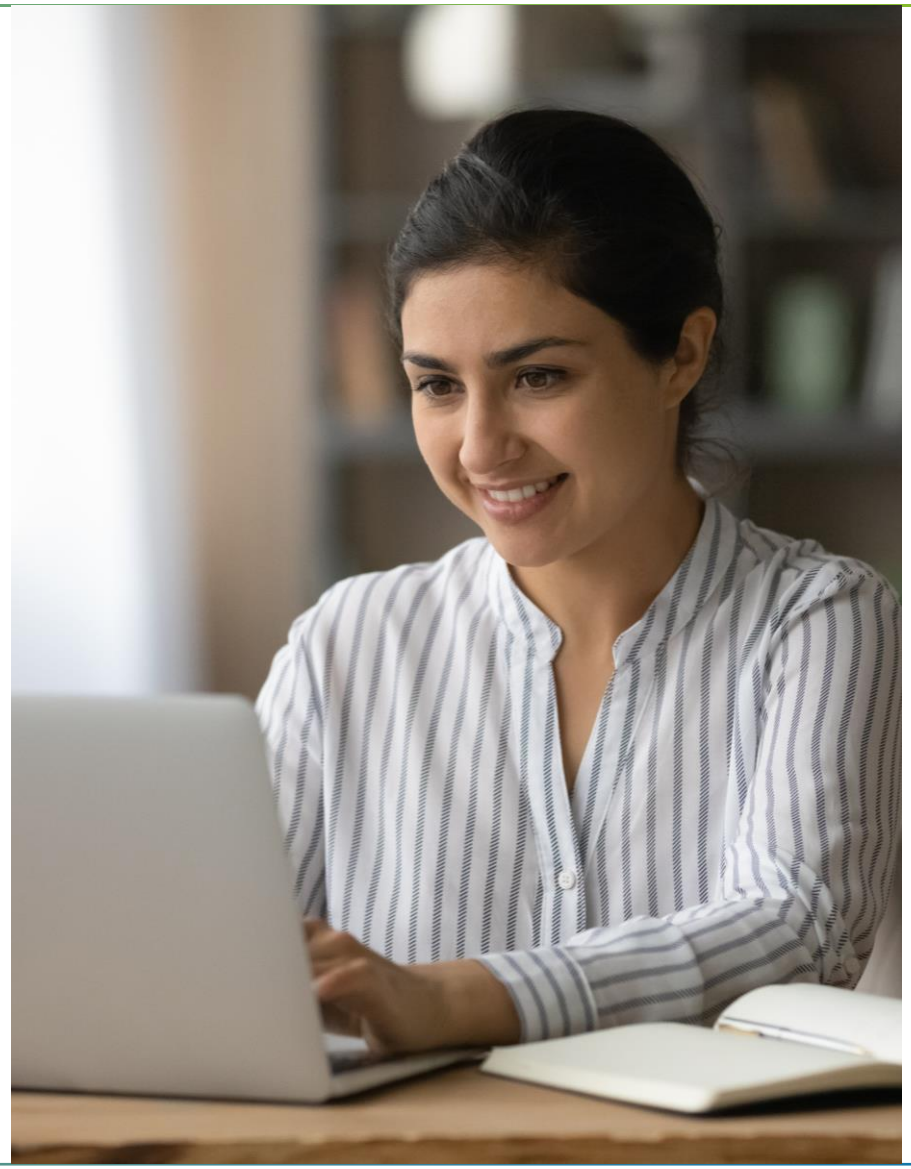
S/4HANA Overview



Journey to S/4HANA Is Inevitable, Why?

S/4HANA enables companies to spearhead towards digital transformation.

- SAP S/4HANA is the digital core for IT transformation
- Respond to rising expectations of customers, employees, and third-parties
- Address the growth of digital data through simplified data architecture
- Enable capabilities for the Internet of Things (IoT) and the digital new economic paradigms
- Develop a competitive edge

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Four Main Areas of Benefits for Customers

Integrated Reporting and Analysis

Real-time information and personalized insights ensure fast and personalized reports and analyses.

New User Experience

Modern role-based user experience based on Systems Applications and Products in Data Processing (SAP) Fiori and enabled by the rich experiences of HTML5.

Application Optimizations

The increased system speed and simplified data model accelerate key processes as batch processing is accelerated tenfold.

Optimized and Scalable Solution

Optimized systems landscape and the choice between different deployment options offer tailored solutions.

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S/4HANA—Who All Does It Benefit?

Moving SAP S/4HANA is two programs in one: a digital transformation program and an Information Technology (IT) re-platforming, so we can have better financial information availability and robust and agile technology.

CEO Perspective: Growth and Insights	• Improve profitability • Grow revenue • Create agility and responsiveness
CIO Perspective: IT Re-Platforming	• Safe and secure transition • Total Cost of Ownership (TCO) improvement • Speed and agility
CFO Perspective: Digital Transformation	• On-the-fly analysis • Single source of truth • Any device and anywhere

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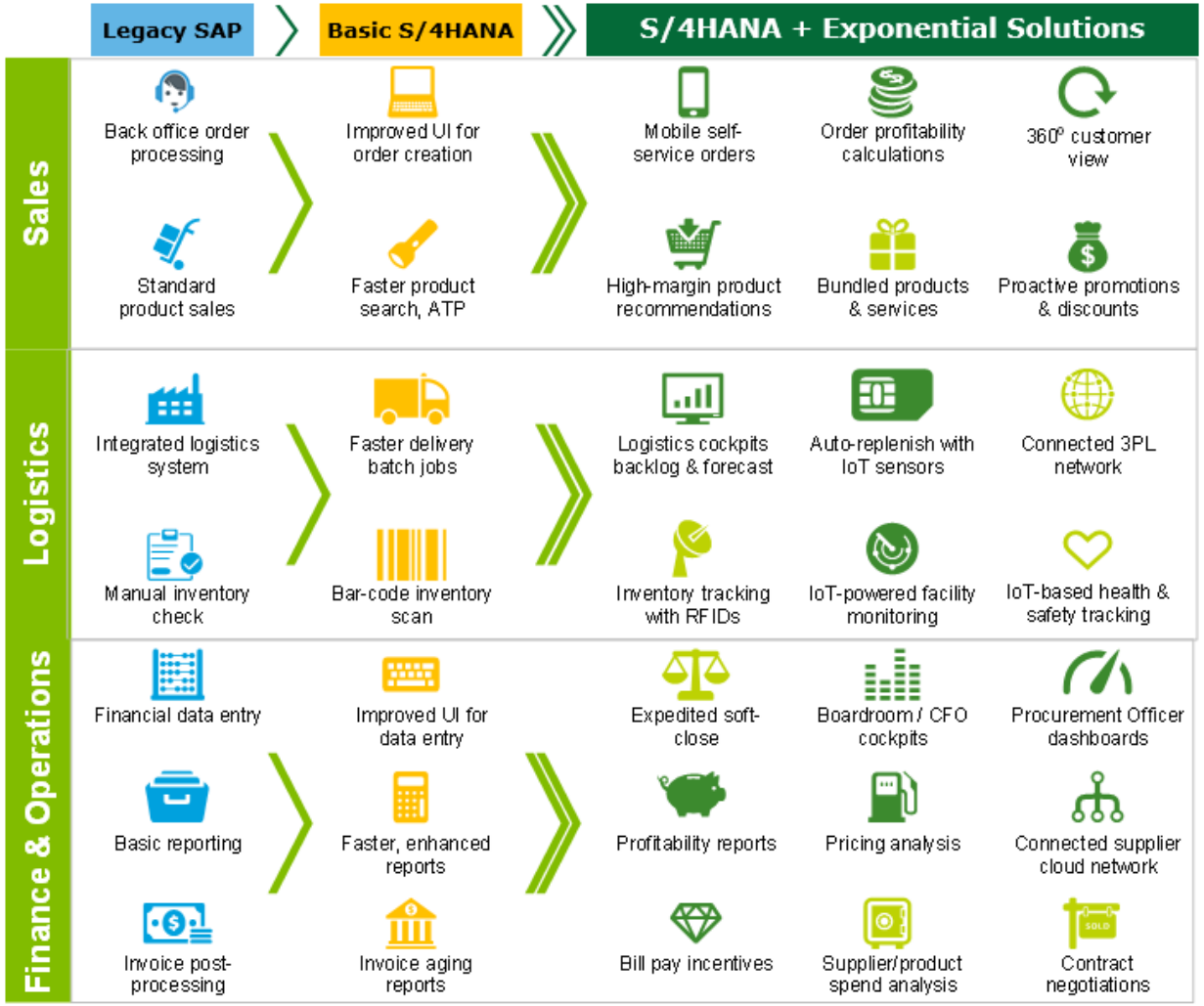
S/4HANA—Who All Does It Benefit? (Cont.)



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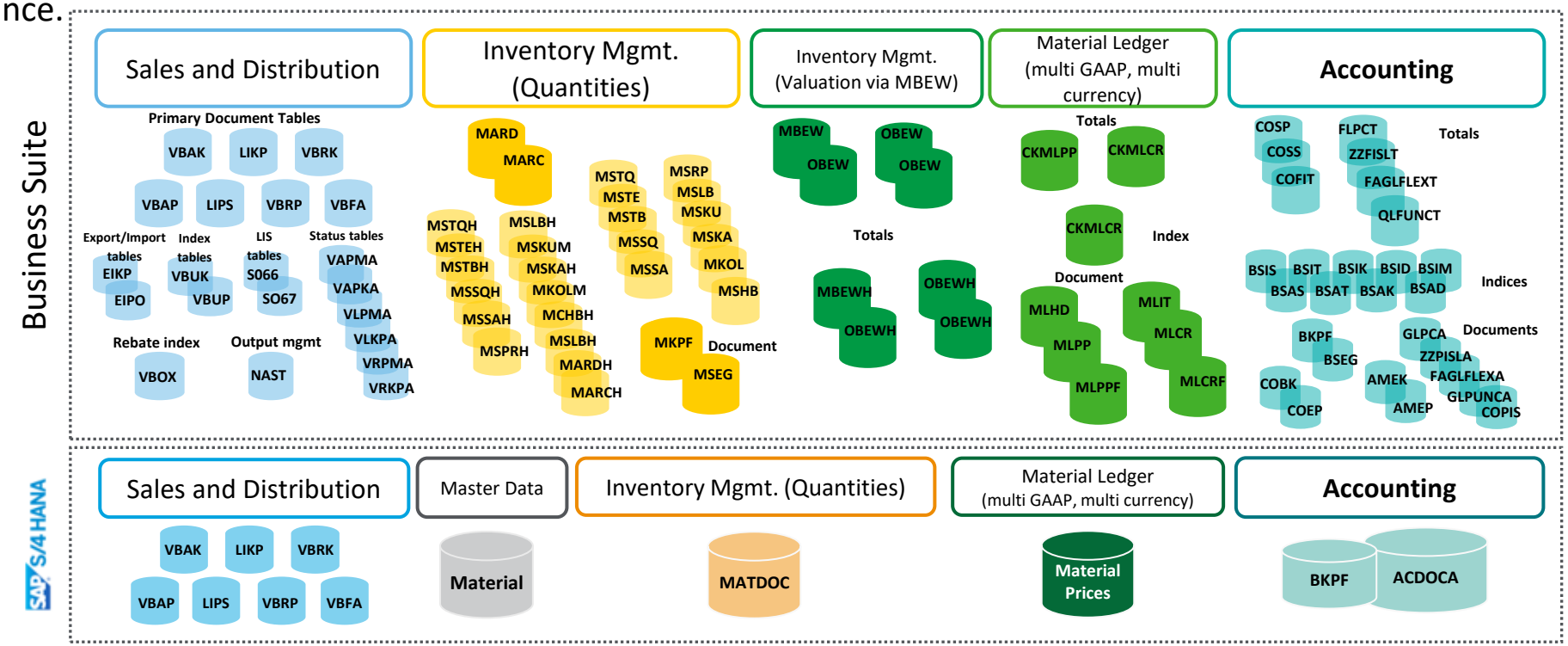
S/4HANA—What Is Data Model Simplification?

Traditional database-based systems have relied on aggregates and indices to overcome performance hurdles leading to redundant and possible inconsistent data storage. S/4HANA enterprise management removes this constraint.

Indices here is nothing but the primary and secondary indices that we saw in the pretraiing. This was done in order to improve the performance.

Take a look at the standard tables like vbak likp etc.



So many tables from the business suite became one or two tables in s/4 hana

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S/4HANA—What Is Data Model Simplification? (Cont.)

< / >

Simple Data Model

- Only main tables remaining, no redundancies
- Clear separation of master data from transactional data
- Reduction of memory footprint

Principle of One

- Only one valuation method (material ledger) instead of 2 (IM + ML)
- Flexible rebate management instead of cumbersome and expensive redundancies

Throughput Increase

INSERT only on database level (MM-IM) no locks for standard price valuation

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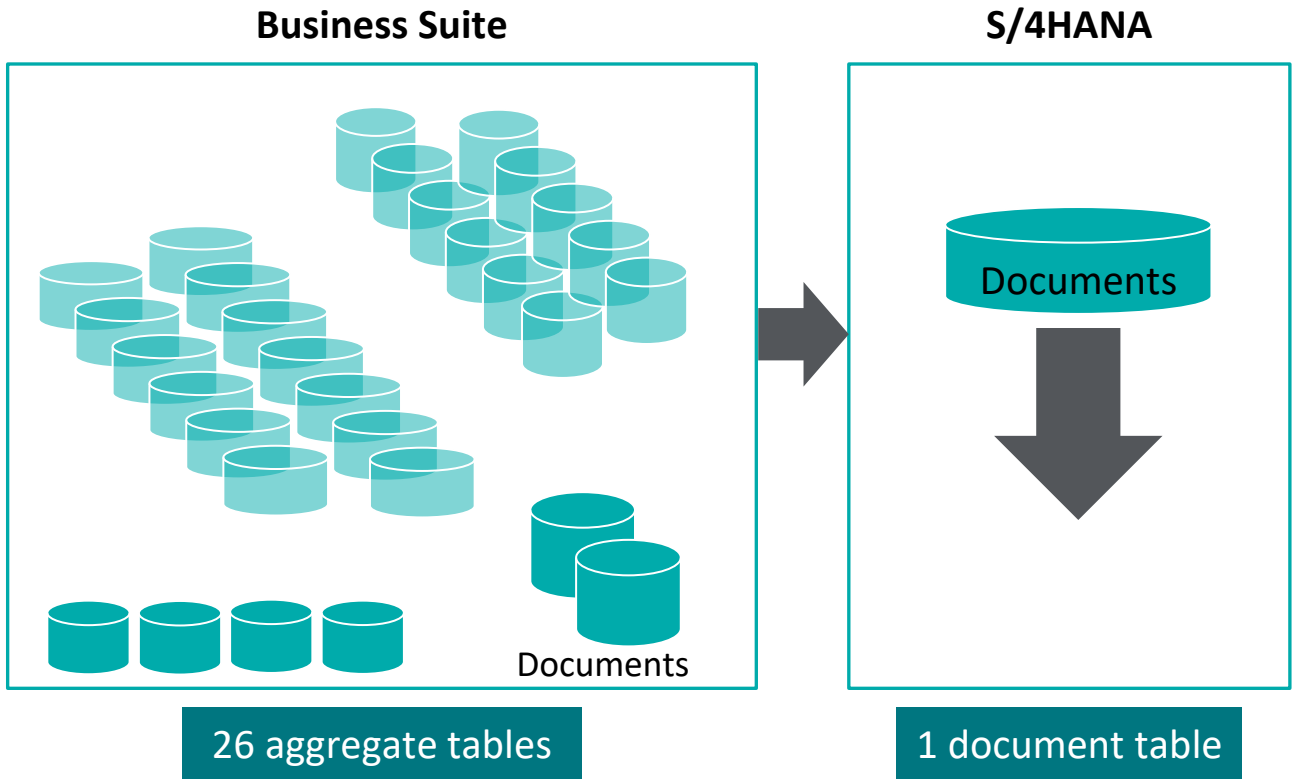
S/4HANA—What Is Data Model Simplification? (Cont.)

Aggregation is now executed dynamically in real time by the HANA database and reverse compatibility is enabled through HANA compatibility views.

- No redundancy reduces storage space, inconsistencies, and confusion.
- Stock is calculated using the history of movements.
- No aggregate tables allowed for no locking and high-performance transactions.
- Legacy applications not optimized for S/4HANA access the new data model through the legacy tables which are now abstracted as views. i.e., Legacy tables become views on the new model and are calculated in real time.



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S/4HANA—User Experience Simplification

SAP ECC traditionally was a transaction-based system where a single user would be required to use several transactions or programs to perform a single task. S/4HANA enterprise management uses Fiori apps and applications focused on users rather than transactions.

- Role-based integrated user experience reduces clicks, inputs, screen changes, and ultimately the duration of activities.
- Fiori is the native user experience for future innovations and will eventually all but replace the SAP Graphical User Interface (GUI). (WebGUI will serve as a bridge until all innovations are available)

User Activity	SAP GUI	SAP Fiori	% Reduction
Duration	2.25 min	1.07 min	53%
Clicks	49	22	55%
Screen changes	26	1	96%
Fields filled	9	3	60%

* Source: https://websmp105.sap-ag.de/~sapidp/012002523100018395812015E.pdf

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What Makes HANA So Unique?

100% In-Memory Computing

Allow Online Transaction Processing (OLTP) and Online Analytical Processing (OLAP) processing in real-time

5-20x Compression

Column-based compression

Up to 10,000x Faster

Massive parallel scaling

Get instant insights into real-time transactions to optimize business operations.

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What Makes HANA So Unique? (Cont.)

Feature	Value
• In-memory columnar store • OLTP + OLAP on a single copy of data • No pre-aggregates and indexes • Run business logic close to the data • Functional libraries • SAP business suite and SAP HANA live	• Value • Faster responses • Insights are on fresh data • Less administration • Faster responses • Simplifies application development • Accelerated business processes

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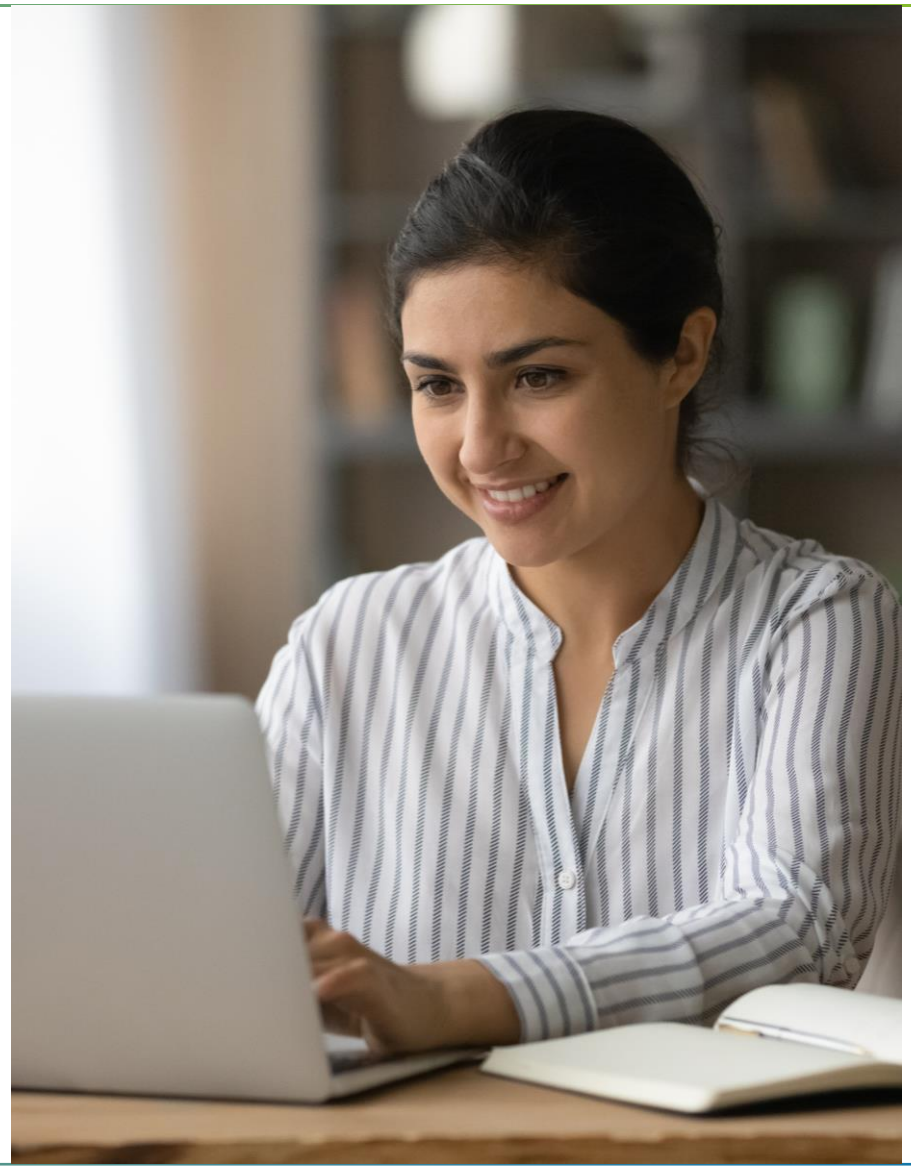
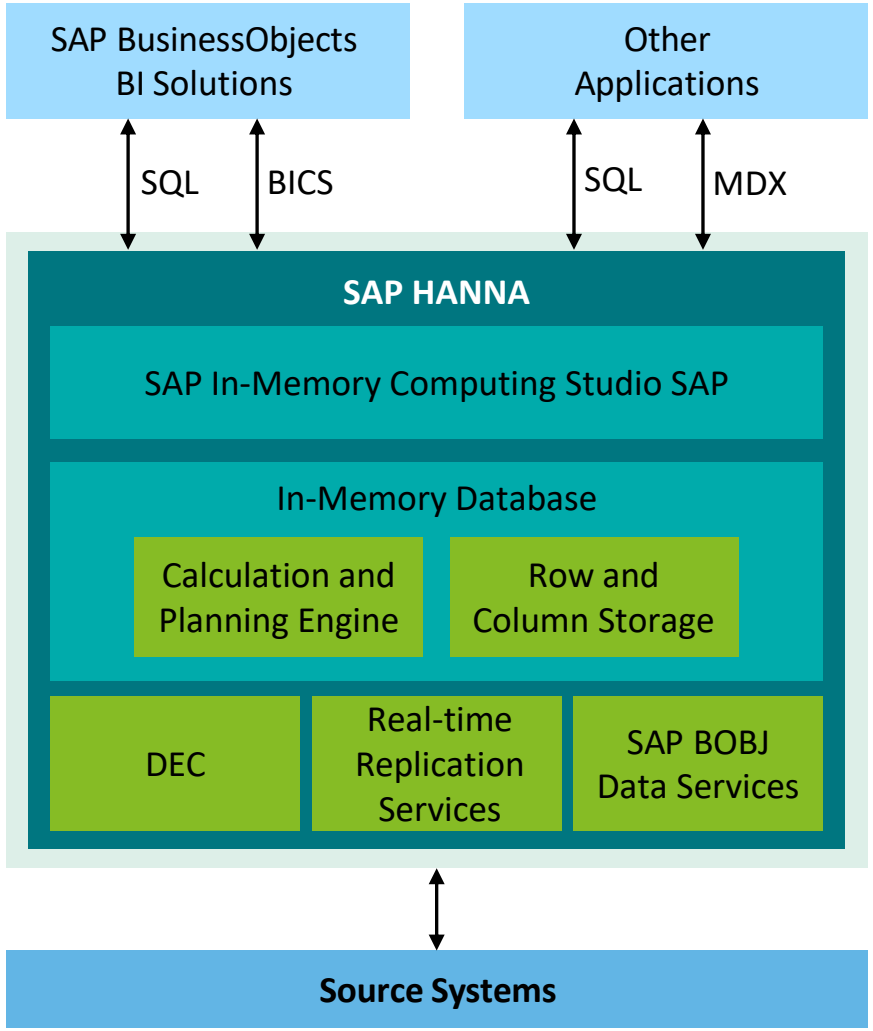
S/4HANA Architecture



SAP HANA High Level Architecture

- A flexible, data source-agnostic in-memory appliance to analyze large volumes of data in real-time.
- Includes tools for data modeling, data and lifecycle management, and security
- Combines SAP software components
- Optimized on hardware (blades)
- Enables to react to business events more quickly through real-time analysis and reporting of operational data
- Non-disruptive to existing landscapes
- Minimizes data duplication and latency

These are analytical tools



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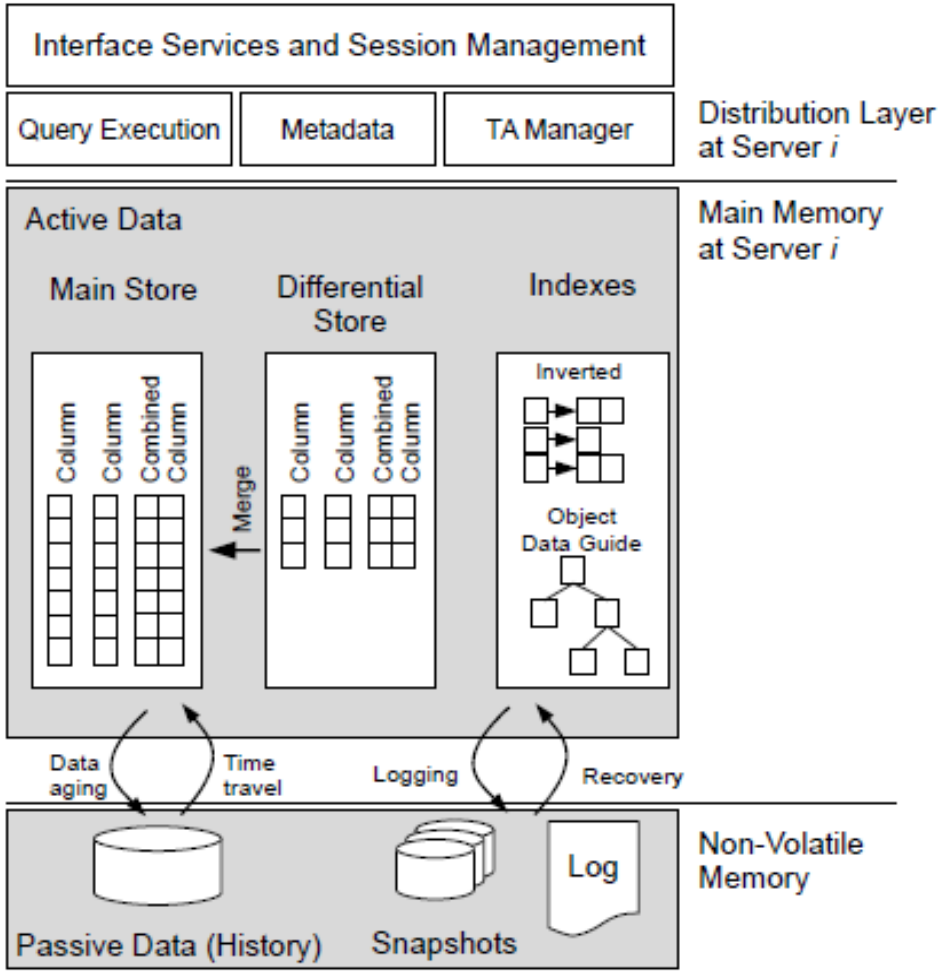
An In-Memory Database for Enterprise Applications

In-Memory Database (IMDB)

- Data resides permanently in the main memory
- Main memory is the primary "persistence“
- Still logging to disk/recovery from disk
- Main memory access is the new bottleneck
- Cache-conscious algorithms/data structures are crucial (locality is king)



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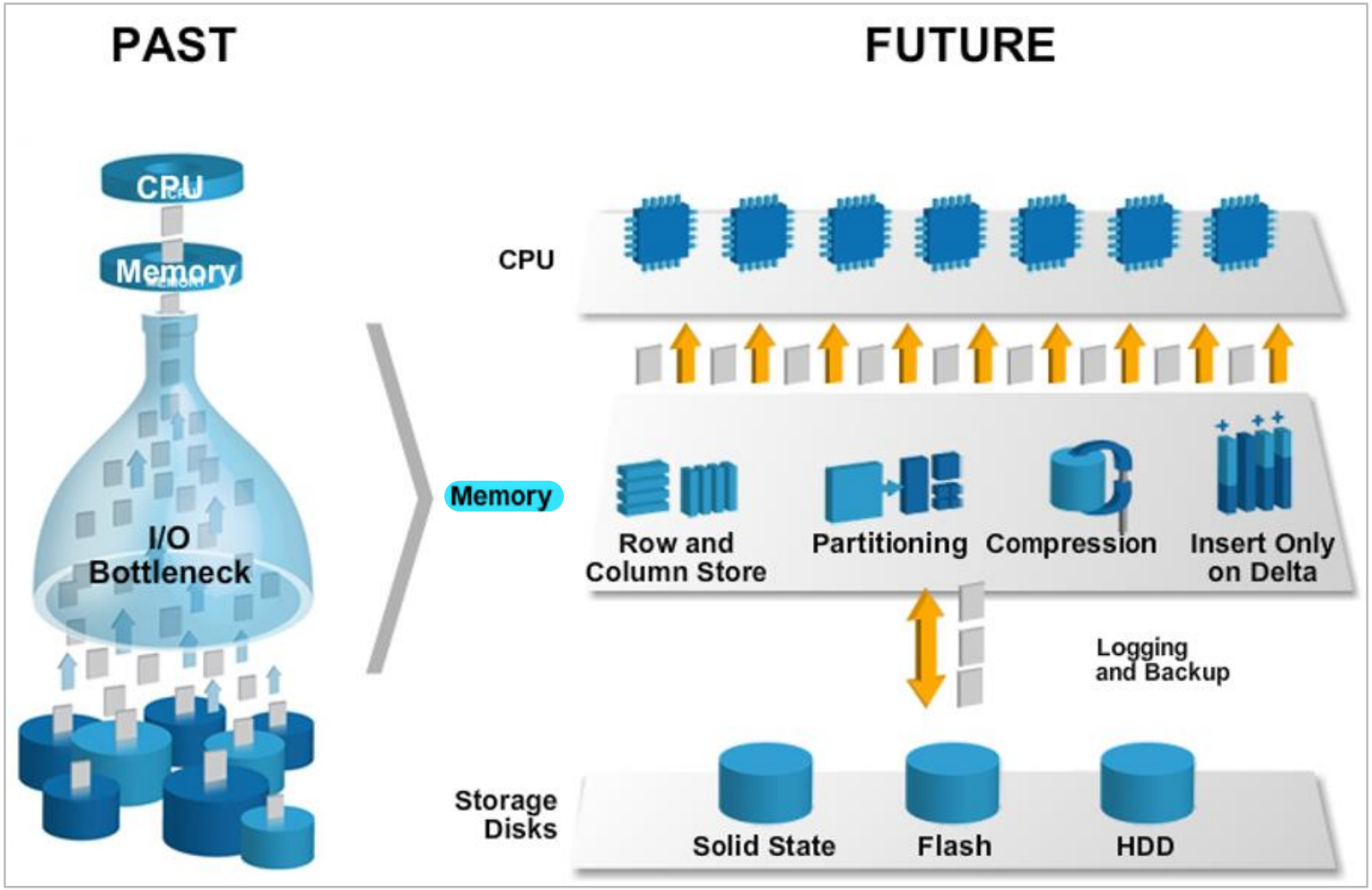
Computer Architecture Is Changing

~~Changes in the hardware technology~~



Some operations are being done in the memory itself. Meaning in memory computing.

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SAP HANA Features at a Glance

Hardware (HW) Technology Innovations

- Multicore architecture (8 x 8 core CPU per blade)
- Massive parallel scaling with many blades
- One blade ~$50,000 = 1 enterprise class server
- 64 bit address space–2 TB in current servers
- 100 GB/s data throughput
- Dramatic decline in price/performance

SAP Software (SW) Technology Innovations

- Row and column store
- Compression
- Partitioning
- No aggregate tables
- Insert only on delta

The convergence of improved hardware economics and technology innovations enables SAP to deliver on its vision of the real-time enterprise with in-memory business applications.

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In-Memory Columnar Store

Faster OLTP + OLAP processing on a single copy of data

- ACID compliant
- High-speed transaction support
- Aggregations on fly
- No indexes for fast access
- Process compressed data
- Optimized for multi-core parallel processing
- Single Instruction, Multiple Data (SIMD) processing support
- Non-Uniform Memory Access (NUMA) optimization to enable future support for very large (20TB+) nodes (Central Processing Unit (CPU)/Memory)
- Schema flexibility: Add columns on the fly

HANA doesn't only use columnar store, it uses hybrid of both column and rows. sotring data in the form of rows is useful in data compression. storing data by column results in better compression than the conventional row sotre.

Need to update all rows, and normally need complex control process

Add columns on the fly: Simple and fast

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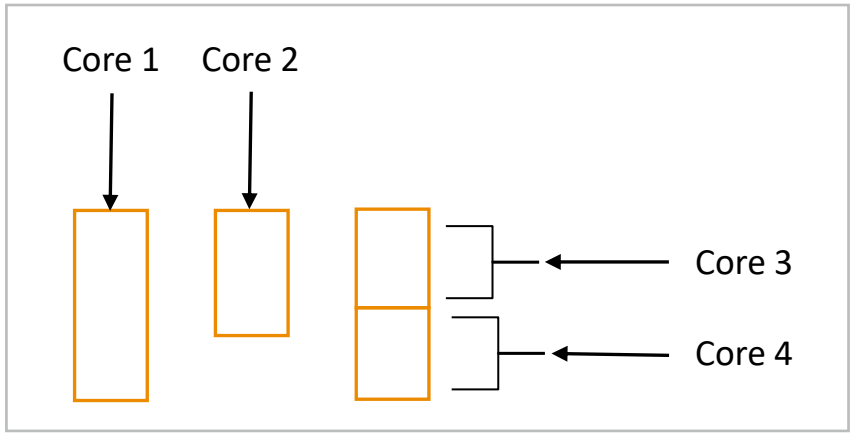
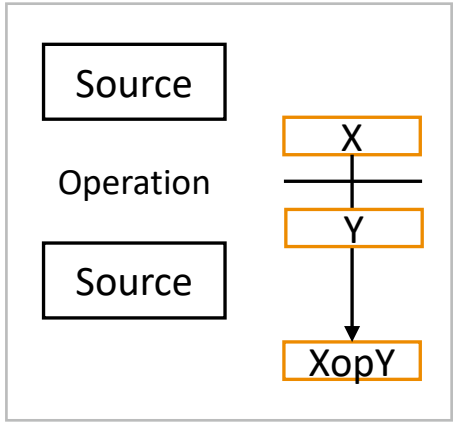
Parallel Processing and SIMD

Each processor can scan a column or portion of a column in parallel.

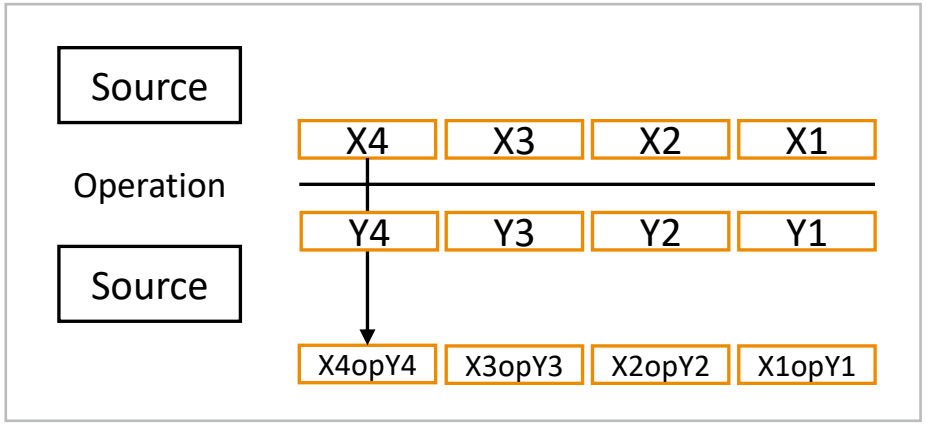


With single instruction, multiple data (SIMD) processing, the same processor operation works on multiple data blocks simultaneously.

Without SIMD



WithSIMD



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Row Store Architecture

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Facts
• Interfaced from the calculation/execution layer • In-memory storage and persistence is managed in the persistence layer • Stores and retrieves data in rows much like a traditional relational database except that the data is stored in memory

Features
• Each row store table has a primary Index, which maps to the ROW ID. ROW ID is part of the primary index of the table • ROW ID specifies the memory segment and page • Indexes only exist in memory • Index definition is stored with the table metadata • Indexes are filled on the fly when the system loads the table into memory

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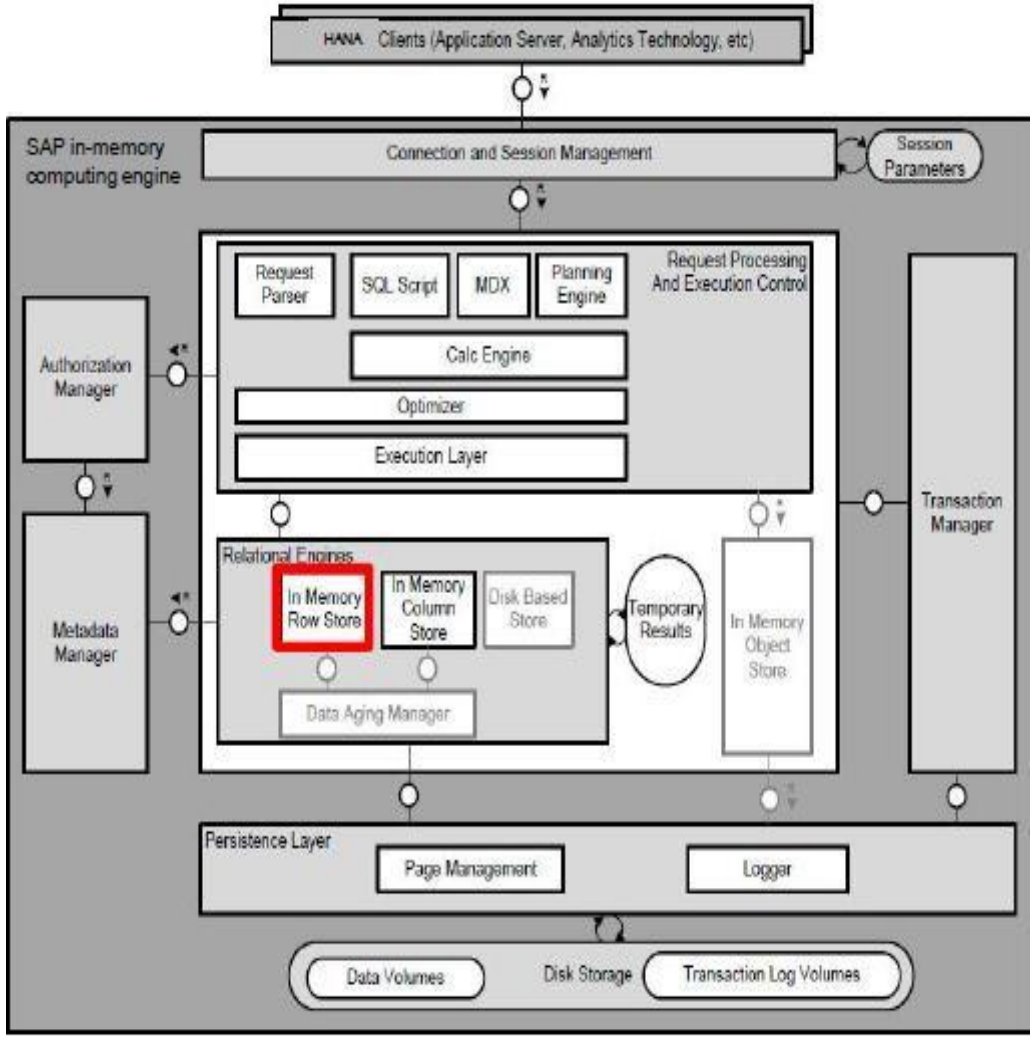
Row Store Architecture (Cont.)



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Row Store Architecture—Components

Transactional Version Memory	Contains temporary versions which are needed for multi-version concurrency control
Segments	Contain actual row store data in pages
Page Manager	The item performs memory allocation and manages free/used pages
Version Memory Consolidation	Garbage collector for Multi Version Concurrency Control (MVCC)
Persistence Layer	Invoked during write operations and performs savepoints

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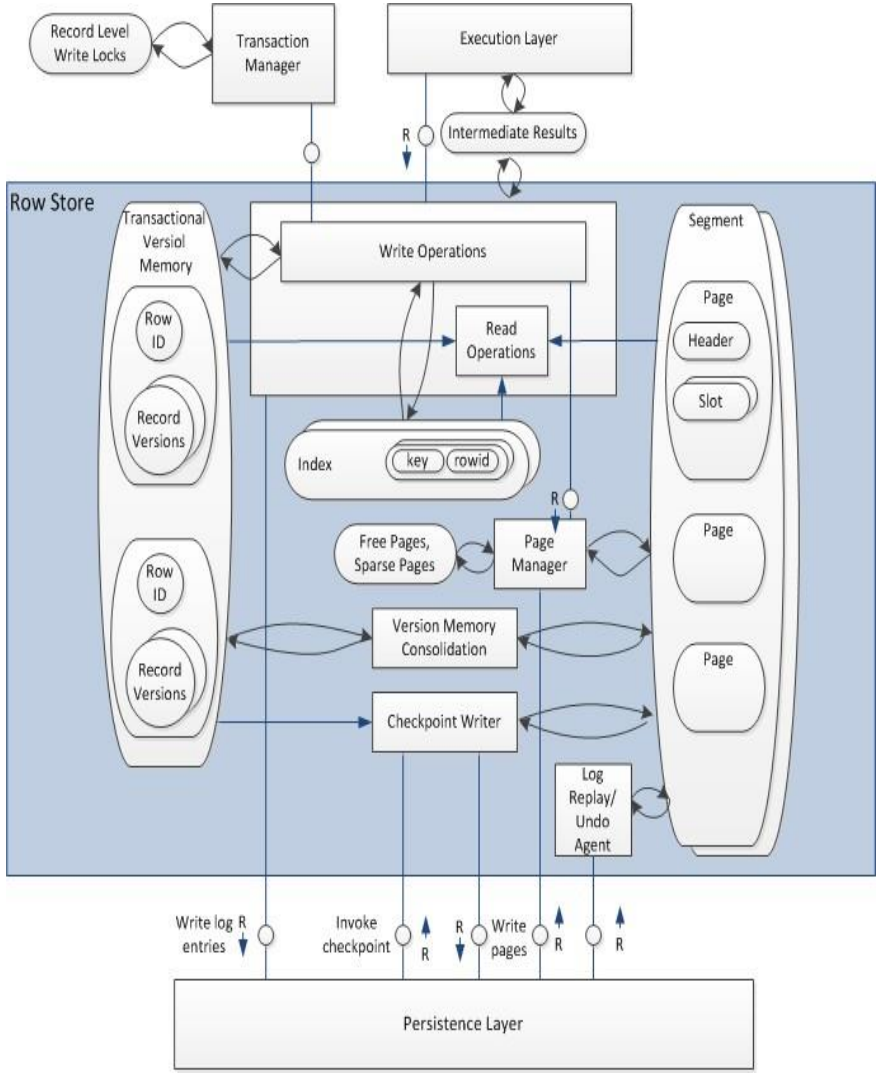
Row Store Architecture—Components (Cont.)



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Row Store Architecture—Highlights

Write Operations

- Mainly goes into “transactional version memory”
- INSERT also write to persisted segment

Persisted Segment Contains

- Visible data
- Committed data

Version Consolidation

- Moves visible version or committed version from transactional version
- Memory into a persisted segment based on commit ID
- Clears outdated record versions from transactional version memory

Memory Handling

- Row store tables are a linked list of memory pages
- Pages are grouped into segments
- Page size: 64 KB

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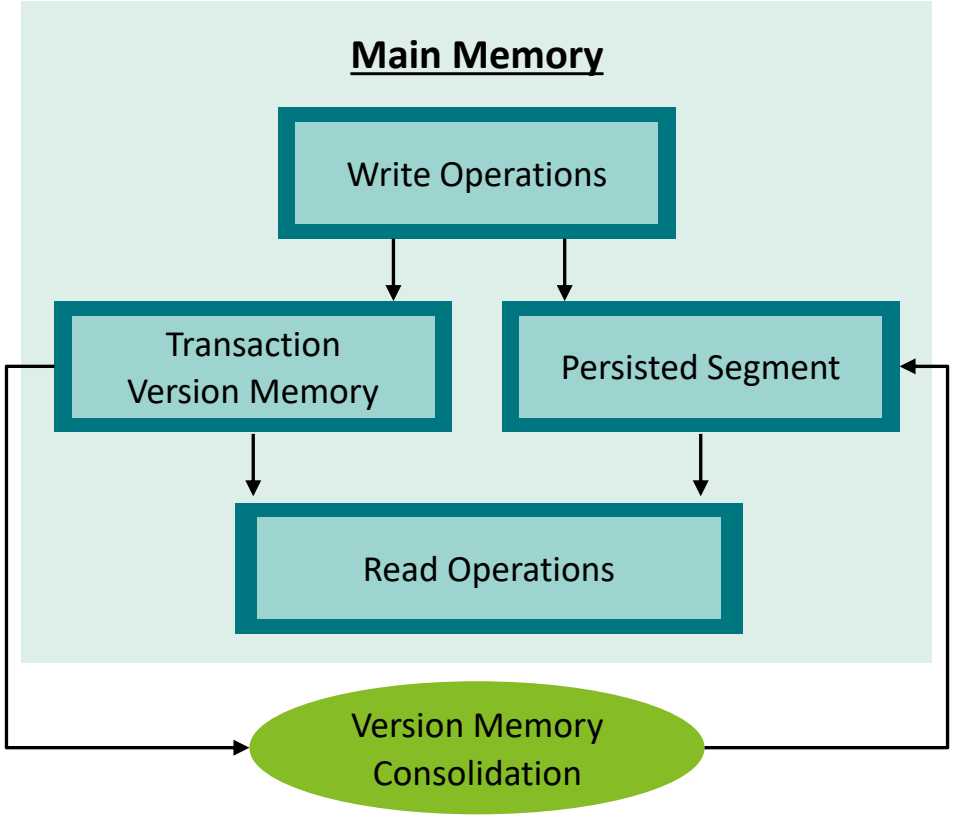
Row Store Architecture—Highlights (Cont.)



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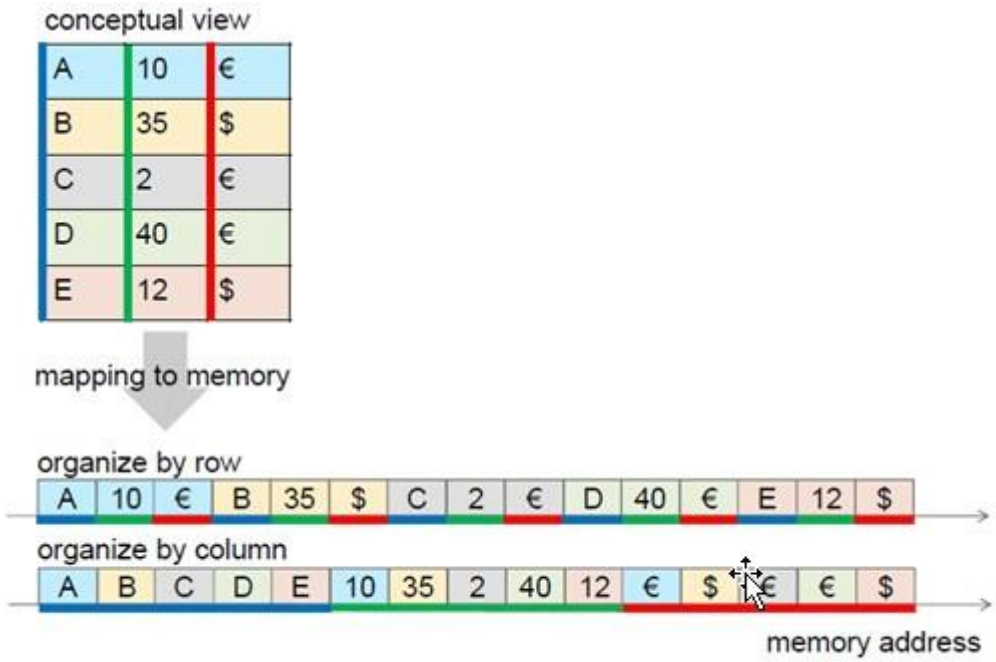
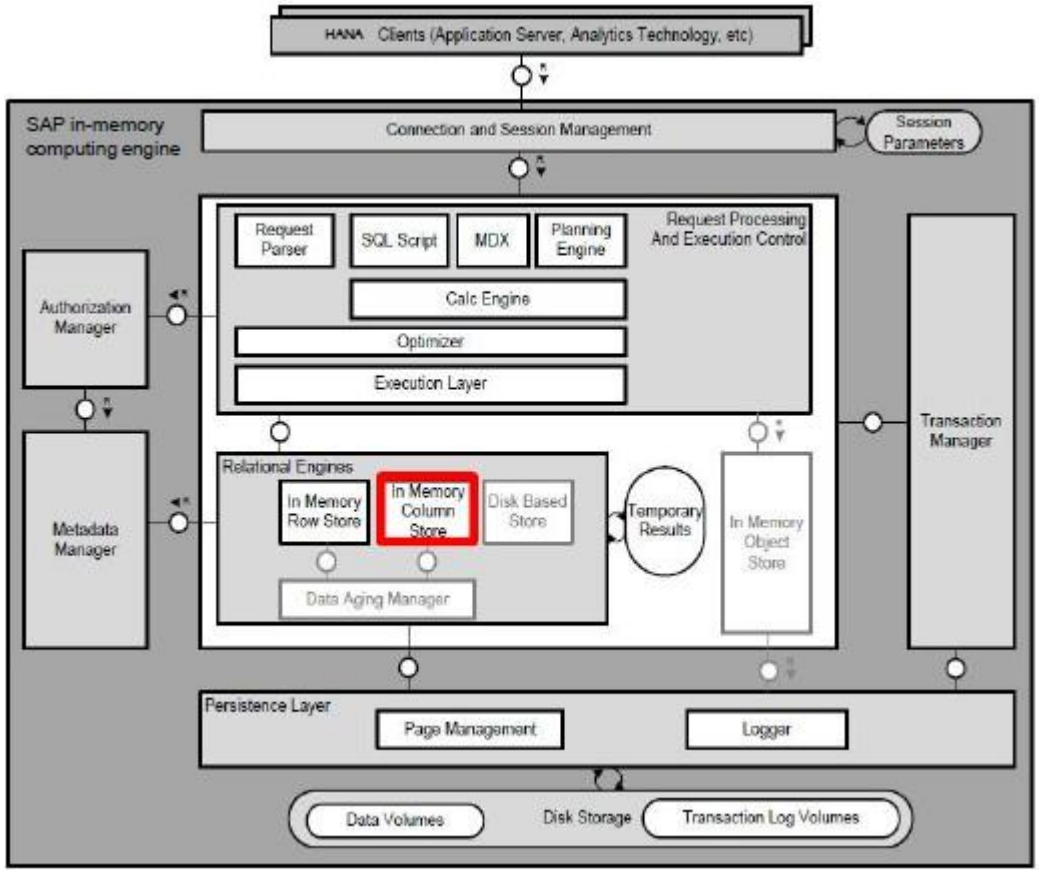
Column Store Data

Facts

- Interfaced from the calculation/execution layer
- In-memory store and persistence are managed in the persistence layer.
- Optimized for reading with efficient data compression





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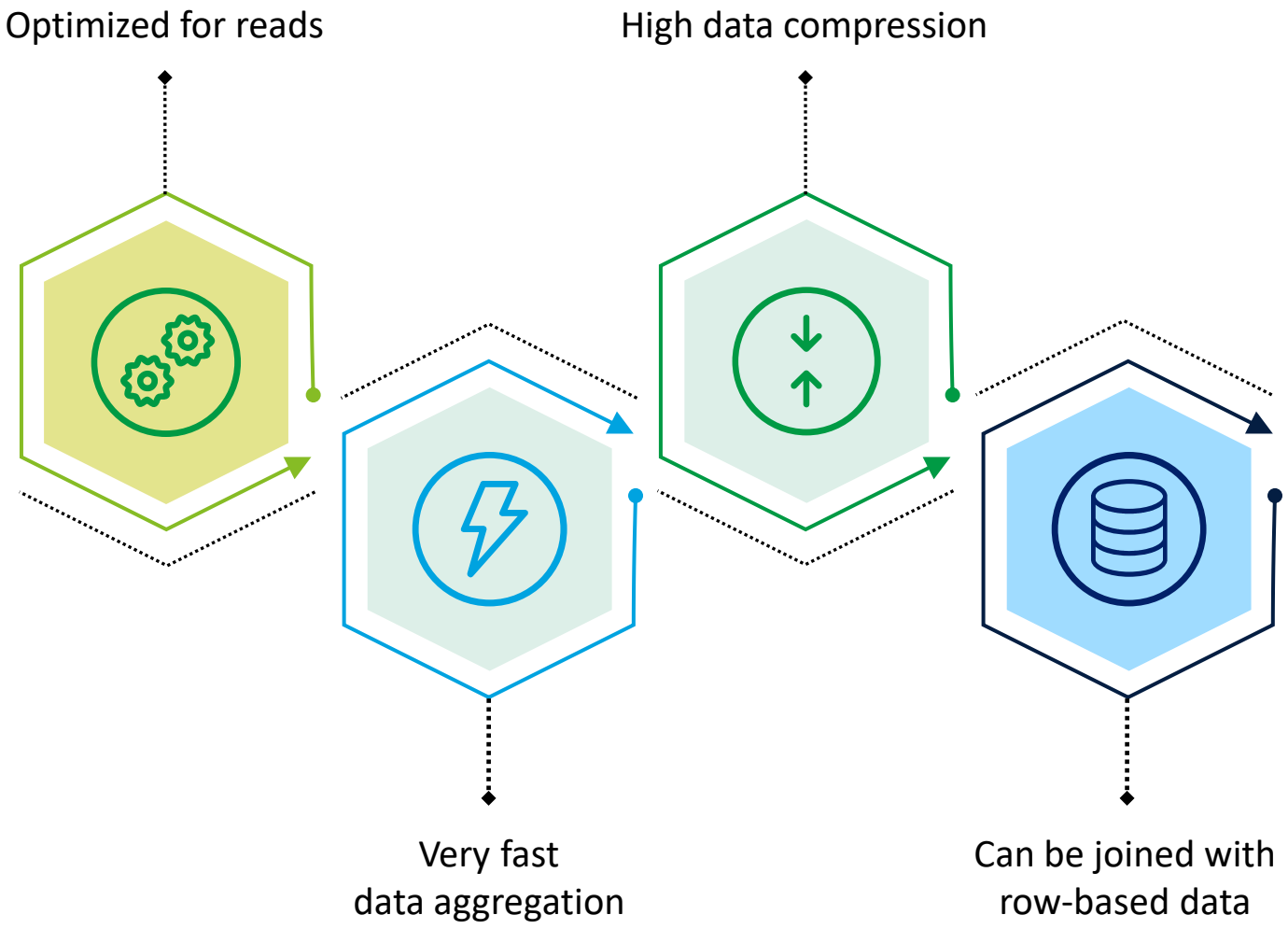
Columnar Data Store Advantages



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Column Store Architecture—Components

Column Store Optimizer and OLAP Optimizer	• Creates physical execution plans • Column store optimizer is cost based and OLAP optimizer is rule based
Main Storage	Contains the main part of the data that is efficiently compressed and read optimized
Delta Storage	Exists only in main memory for storing changes
History Storage	Contains data versions which are used in MVCC
Load/Unload Agent	Used for loading data from persistence layer, which is preloaded on startup or loaded on demand

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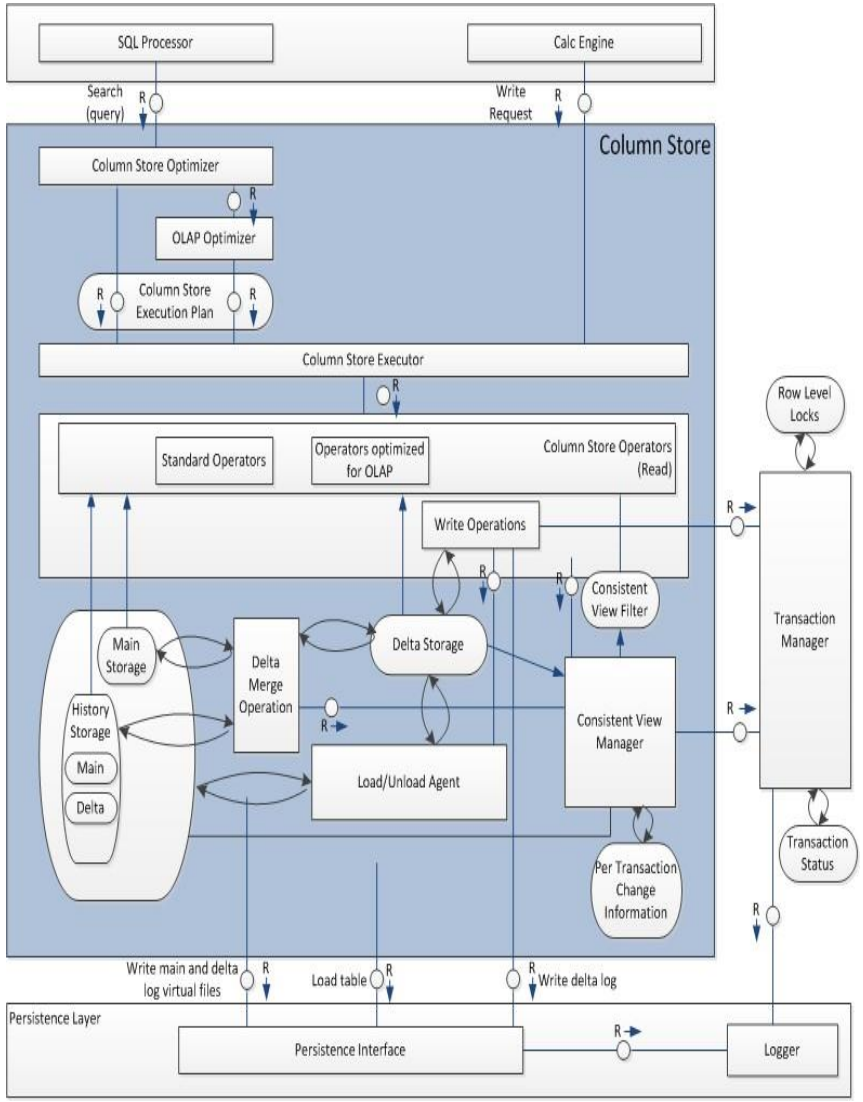
Column Store Architecture—Components (Cont.)



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Column Store Architecture—Highlights

Storage Separation

(Main and Delta)

Enables high compression and high write performance at the same time

Write Operation

- Only in the delta storage because write optimized
- The update is performed by inserting a new entry into the delta storage

Read Operation

- Always have to read from main and delta storages and merge the results
- In-Memory Computing Engine (IMCE) uses MVCC to ensure consistently read operations

Delta Merge Operation

To move changes in delta storage into the main storage because main storage is read optimized happened asynchronously

Modeling is only possible for columns-based tables. (HANA) System tables are created where they fit best.

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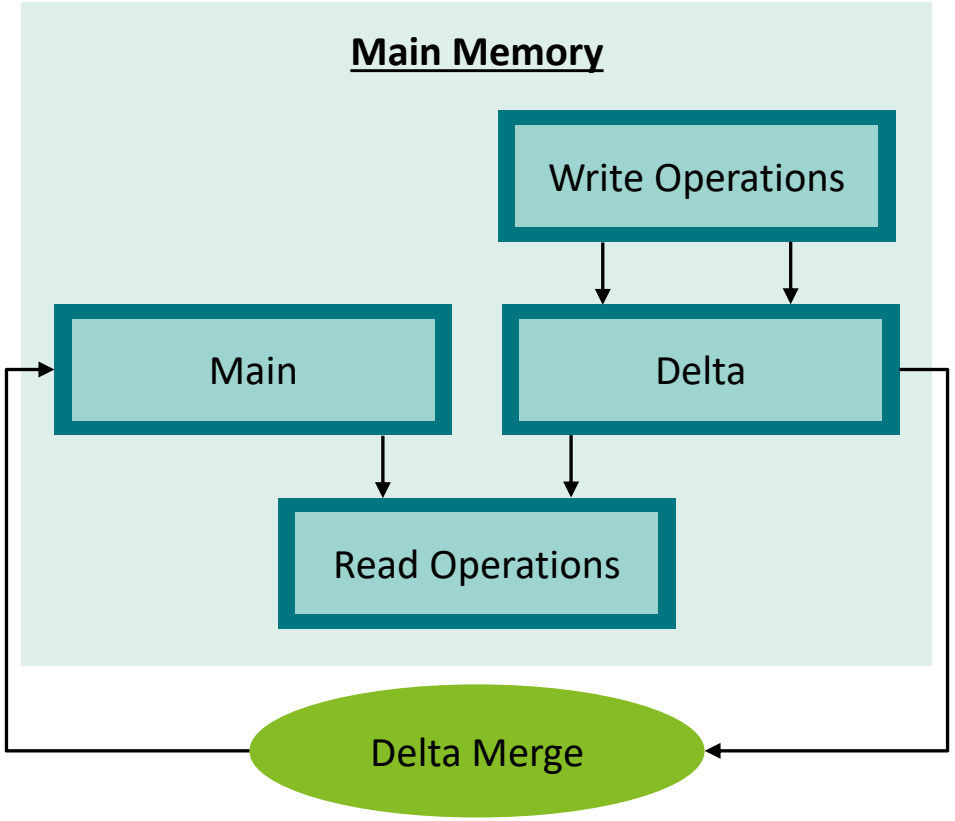
Column Store Architecture—Highlights (Cont.)



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Multi Version Concurrency Control (MVCC)

- The SAP HANA database provides full atomicity, consistency, isolation, and durability (ACID) transactions.
- For concurrency control, the SAP HANA database implements the classical MVCC principle that allows long-running read transactions without blocking update transactions.
- MVCC in combination with a time-travel mechanism, allows temporal queries inside the relational engine.
- With the MVCC concurrency control model, locks acquired for querying (reading) data do not conflict with locks acquired for writing data, hence reading never blocks writing, and writing never blocks reading.

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Key Change Areas

- Business Partner
- Output Management
- Financial Closing Cockpit

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Key Change Areas (Cont.)

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Business Partner (BP)

- Creation/migration of vendors, customers, employees, and contact persons as business partners.
- Conversion programs provided by SAP for migration activity.
- Tcodes like XK01/XD01 replaced by BP. (Security roles need to be adjusted)

Output Management

- Output management for billing and purchase orders moved to Business Rule Framework plus (BRF+) from NAST.
- Output through print, email, fax, and intermediate document (Idoc) are supported by BRF+.
- Outputs for old documents will still be governed by NAST.

Financial Closing Cockpit

- Financial closing cockpit can now launch in the web application. (SAP NetWeaver business client)
- Cross Template Task Monitor– Tasks assigned in multiple templates can be monitored centrally.
- Old Closing cockpit still works (not supported by SAP going forward).

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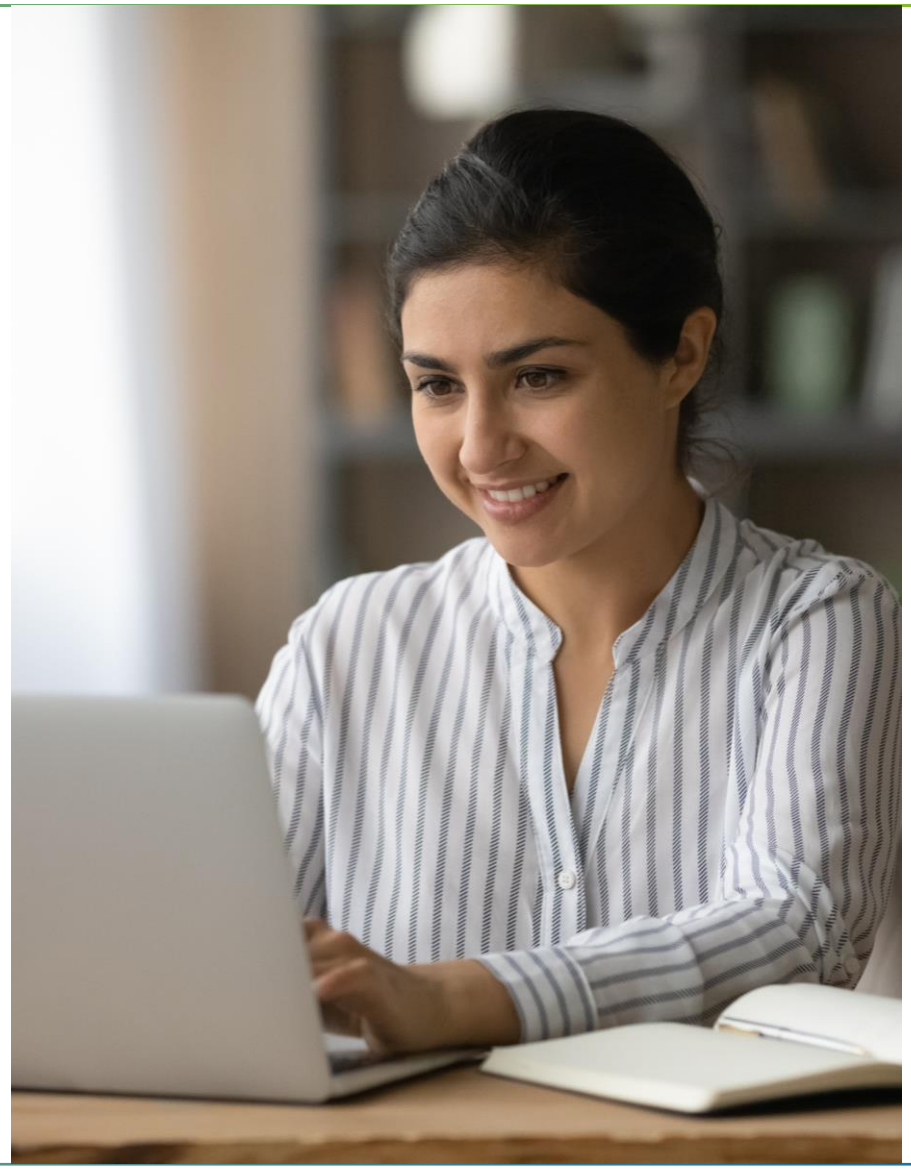
S/4HANA Migration



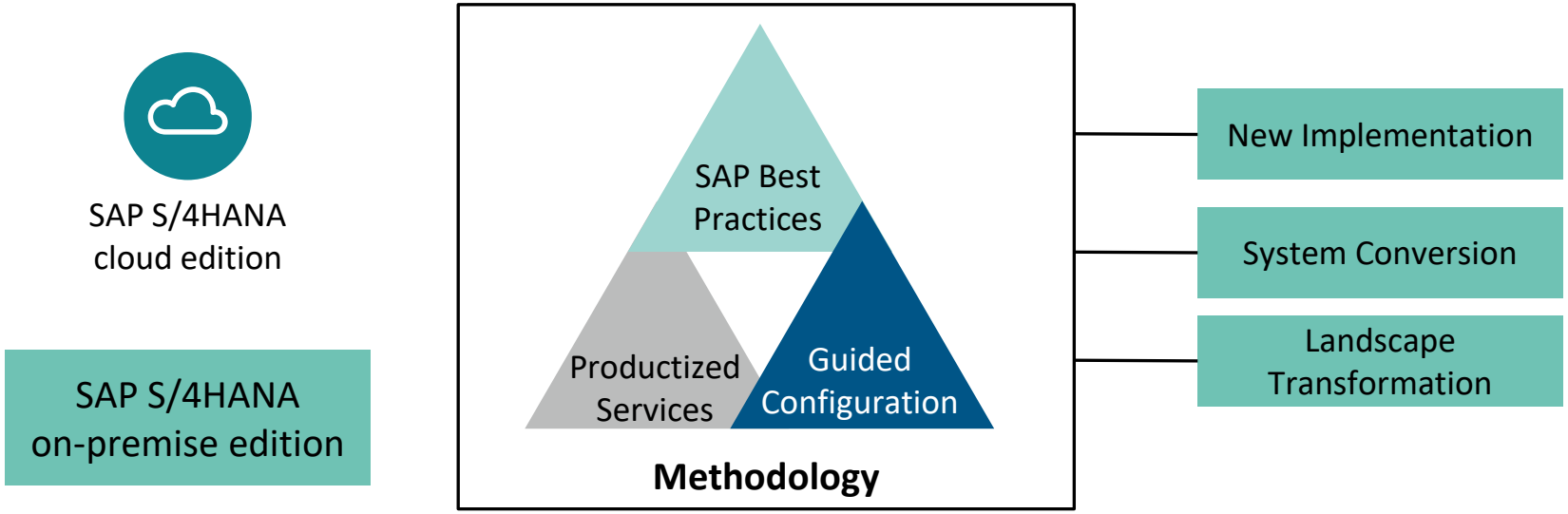
S/4HANA Enterprise Management—SAP Activate

S/4HANA enterprise management introduces a new accelerator product titled SAP activate that looks to reduce the time and effort needed to implement S/4HANA enterprise management.

A technology enabled implementation/conversion methodology built into S/4HANA.

- Combines SAP's best practices into a single package including template organization, data, and basic scenarios
- Guided configuration for setup of simple scenarios
- Test scenarios for validating the solution

Evolution of SAP's classical Accelerated SAP (ASAP) methodology with a focus on the cloud but applicability for on-premises.



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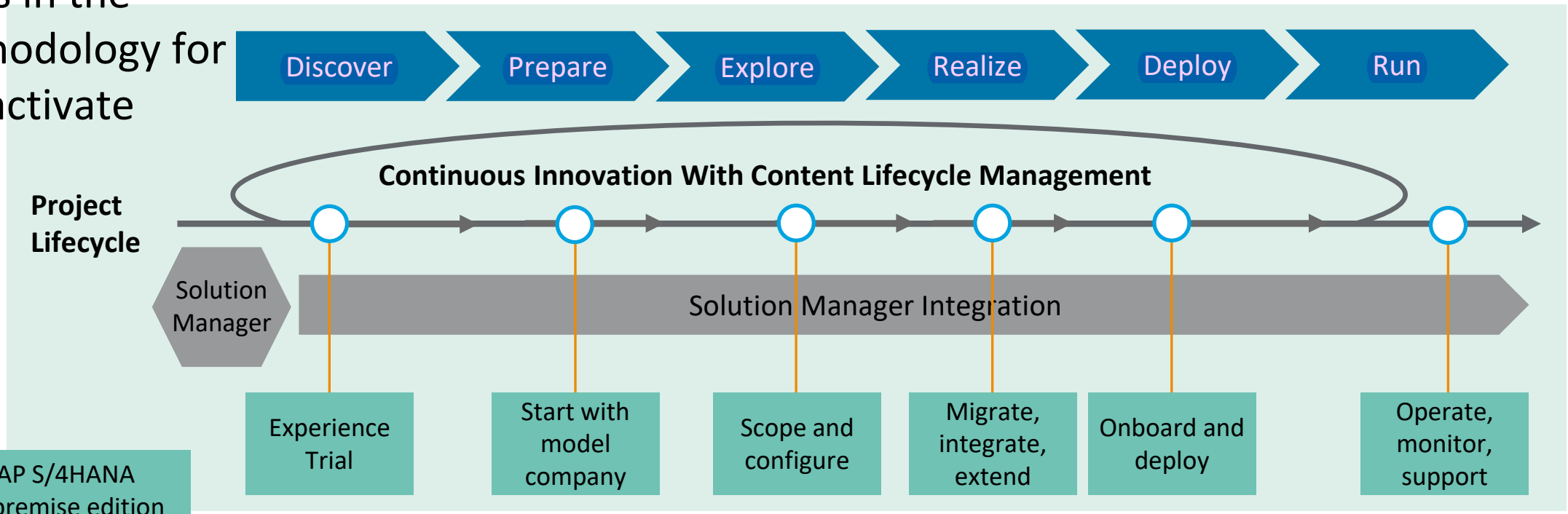
S/4HANA Enterprise Management—SAP Activate (Cont.)

Steps in the methodology for sap activate



SAP S/4HANA on-premise edition

SAP S/4HANA cloud edition

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Transitioning to S/4HANA Enterprise Management

There are three main scenarios where a client may start when transitioning to S/4HANA Enterprise Management.

New Implementation	ERP System Non-SAP System	new implementation is also called green field implementation	SAP S/4HANA on-premise edition SAP S/4HANA cloud edition	New or existing SAP customers implementing a new SAP S/RHANA system with an initial data load
Brownfield/ System Conversion	ERP System		SAP S/4HANA on-premise edition	Complete conversion of an existing SAP business suite system to SAP S/4HANA
Landscape Transformation	ERP System - Region A - ERP System - Region B - ERP System - Region C -		SAP S/4HANA on-premise edition SAP S/4HANA cloud edition	Consolidation of the current regional SAP business suite landscape into one global SAP S/4HANA system

Both new implementations and landscape transformations are fundamentally the same processes; while system conversion is unique.

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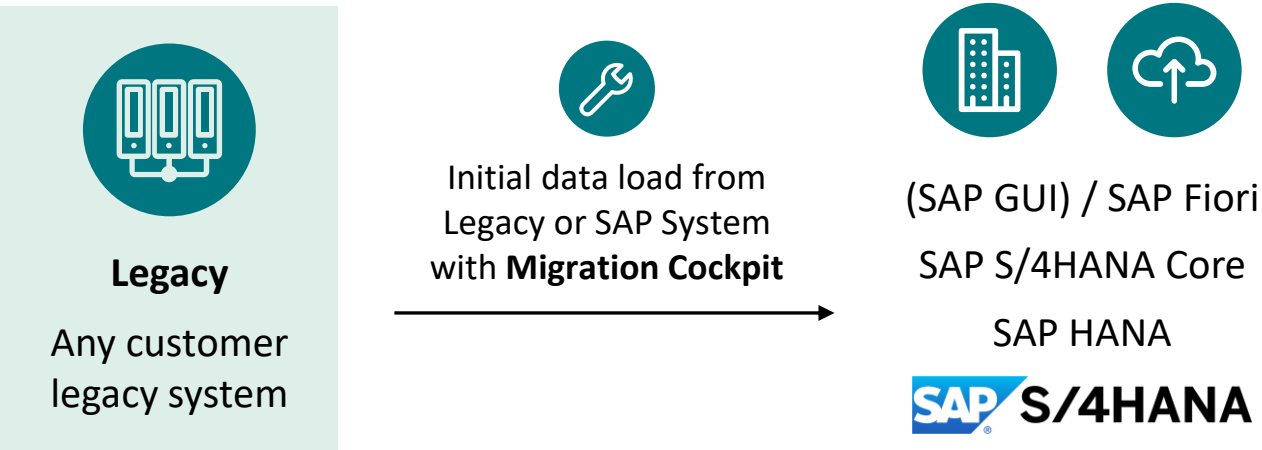
SAP S/4HANA—Transition

Scenario 1—New Implementation

Scenario Description: New implementation of SAP S/4HANA

e.g., for customers migrating to a legacy system

What?	How?
Install SAP S/4HANA	Software Provisioning Manager (SWPM)
Initial data load from source system	• SAP S/4HANA migration cockpit (legacy system via file upload) • Optional: SAP data services with SAP best practice migration content



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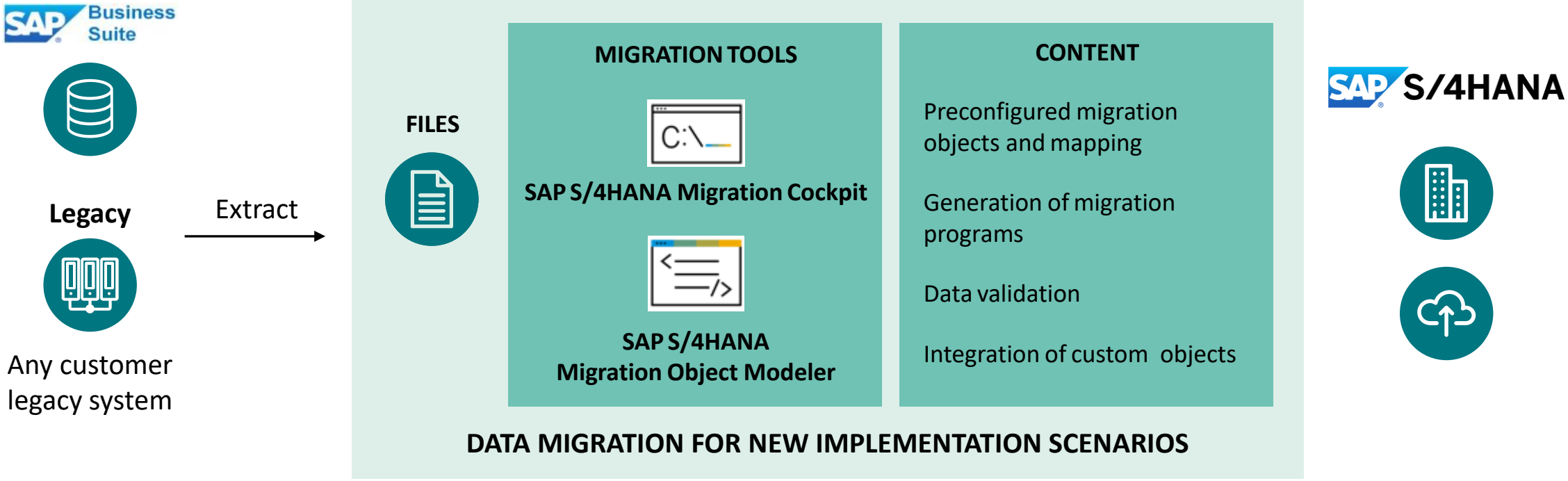
SAP S/4HANA Migration Cockpit—Overview



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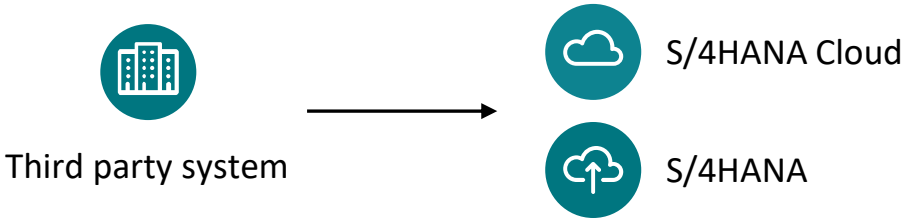
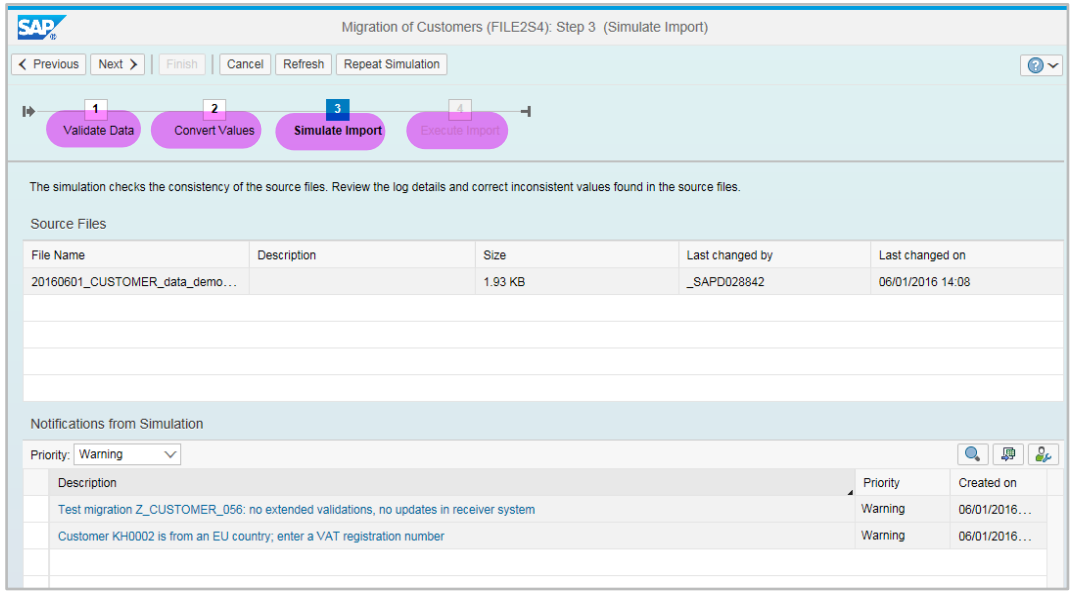
SAP S/4HANA Migration Cockpit

Speeding Time to Value

The next-generation data migration software embedded in S/4

- Shorten data migration project duration by more than 20%
- Highly simplified for SAP applications as data sources
- Guided procedure takes users through the migration process: Easy and safe
- No programming by the customer required
- Flexible for integration of custom business data including data transformations
- Included in S/4HANA license and shipment
- Made for migrating data from SAP and non-SAP sources into SAP S/4HANA (on-premises) and SAP S/4HANA Cloud





Step-by-step guidance through the migration process

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SAP S/4HANA Migration Object Modeler

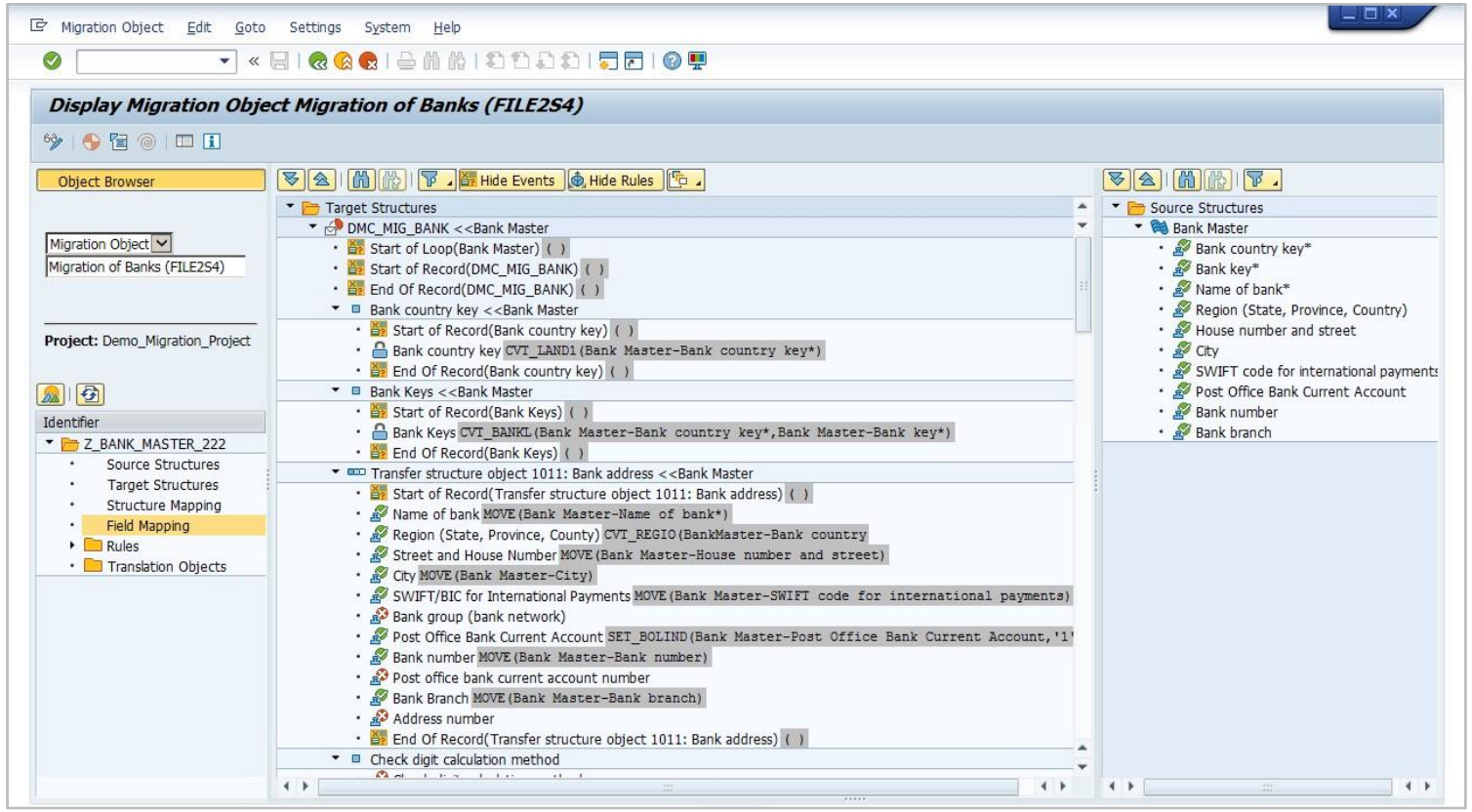
The SAP S/4HANA migration object modeler is part of the SAP S/4HANA Migration cockpit.

A design tool to easily integrate custom objects and enhancements

Integration of newly created objects

- Custom objects
- SAP standard objects that are not yet in the scope of the SAP S/4HANA migration cockpit

Customer-specific enhancements (e.g., adding new fields) to SAP standard objects delivered with the SAP S/4HANA migration cockpit.



[Follow this link to SAP Help for the SAP S/4HANA Migration Cockpit]

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SAP S/4HANA—Transition

Scenario 2—System Conversion

Scenario Description:

A complete technical in-place conversion of an existing SAP business suite Enterprise Resource Planning (ERP) system to SAP S/4HANA (Database, SAP NetWeaver, and application transition).

Why Would You Choose This Option?

- Bring your existing business processes to the new platform.
- Keep your investments in custom-specific processes and developments.
- Keep your complete data.
- Mitigate the risk and investment of a big bang conversion project by reducing the scope of the transition project to a pure technical conversion project and adopting new innovations at your speed at a later point in time and in a phased approach.

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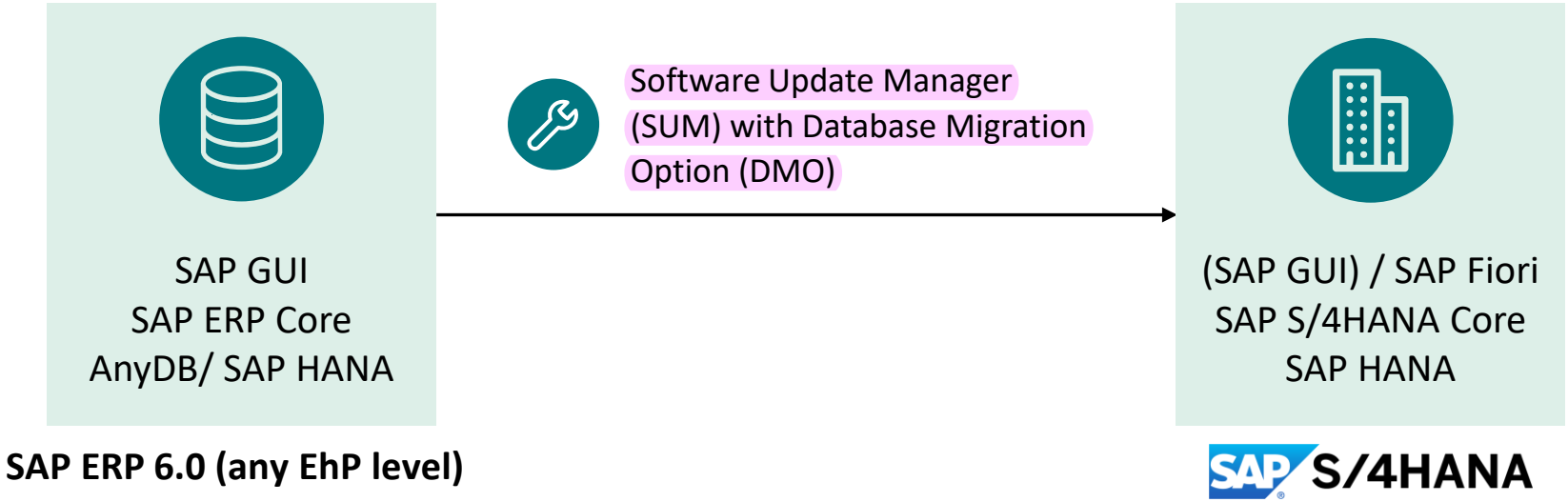
SAP S/4HANA—Transition (Cont.)



What?	How?
In-place technical conversion	Tool-based technical conversion process of SAP business suite ERP system to SAP S/4HANA using SAP Software Update Manager (SUM); Downtime minimizing options available

SCN Blog: SAP S/4HANA System Conversion – At a glance

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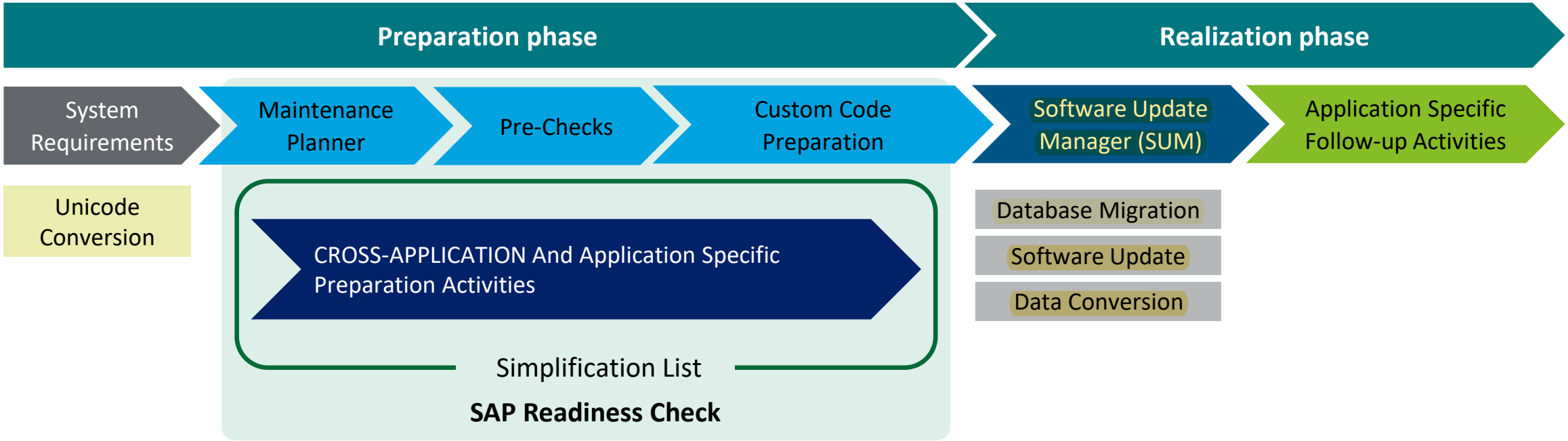
SAP S/4HANA System Conversion

Overview Technical Process Steps



Simplification List for SAP S/4HANA

To allow our customers better planning and estimation of their way to SAP S/4HANA, a “Simplification list for SAP S/4HANA” is created. This list it is described in detail on a functional level what happens in S/4HANA to individual transactions and solution capabilities. In some cases, SAP has merged certain functionality with other elements or reflected it within a new solution/architecture compared to the SAP business suite products.

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Maintenance Planner

</>

- The maintenance planner is a solution hosted in the SAP support portal at: http://apps.support.sap.com/sap/support/mp.
- It simplifies the landscape maintenance process to update, upgrade, and install systems with ease and efficiency by checking the system with regard to business functions, industry solutions, and add-ons.
- If there is no valid path for the conversion (for example, the add-on is not released yet), the maintenance planner prevents the conversion.
- The maintenance planner is a pre-requisite for the system conversion to SAP S/4HANA. The Maintenance Optimizer (MOPZ) is not used. Entry via the 'plan for SAP S/4HANA' entry tile.

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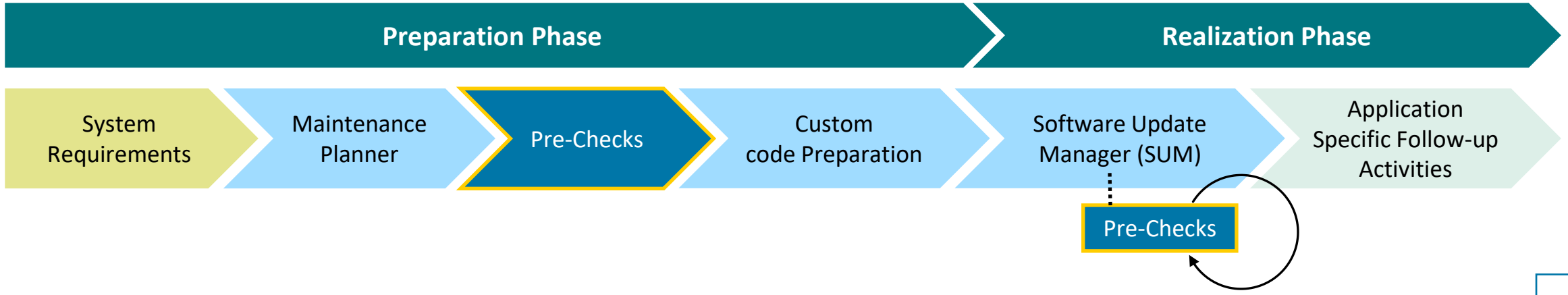
Pre-Checks

- Pre-checks are shipped as SAP notes to customers that want to convert to SAP S/4HANA. Customers can use these pre-checks to find out what mandatory steps they have to carry out before converting to S/4HANA.
- They are executed on the (SAP business suite) start system (preparation phase) to determine relevant simplification items. The results list the simplification items that need addressing before attempting the conversion process.
- They are executed again as a step in the Software Update Manager (SUM). In case dedicated simplification items require mandatory preparation steps, the conversion procedure can stop to ensure a consistent system conversion. The conversion is stopped if errors persist.

For any changes in the opeations of standard objects SAP releaes notes. the tcode for the sap notes it SNOTE.



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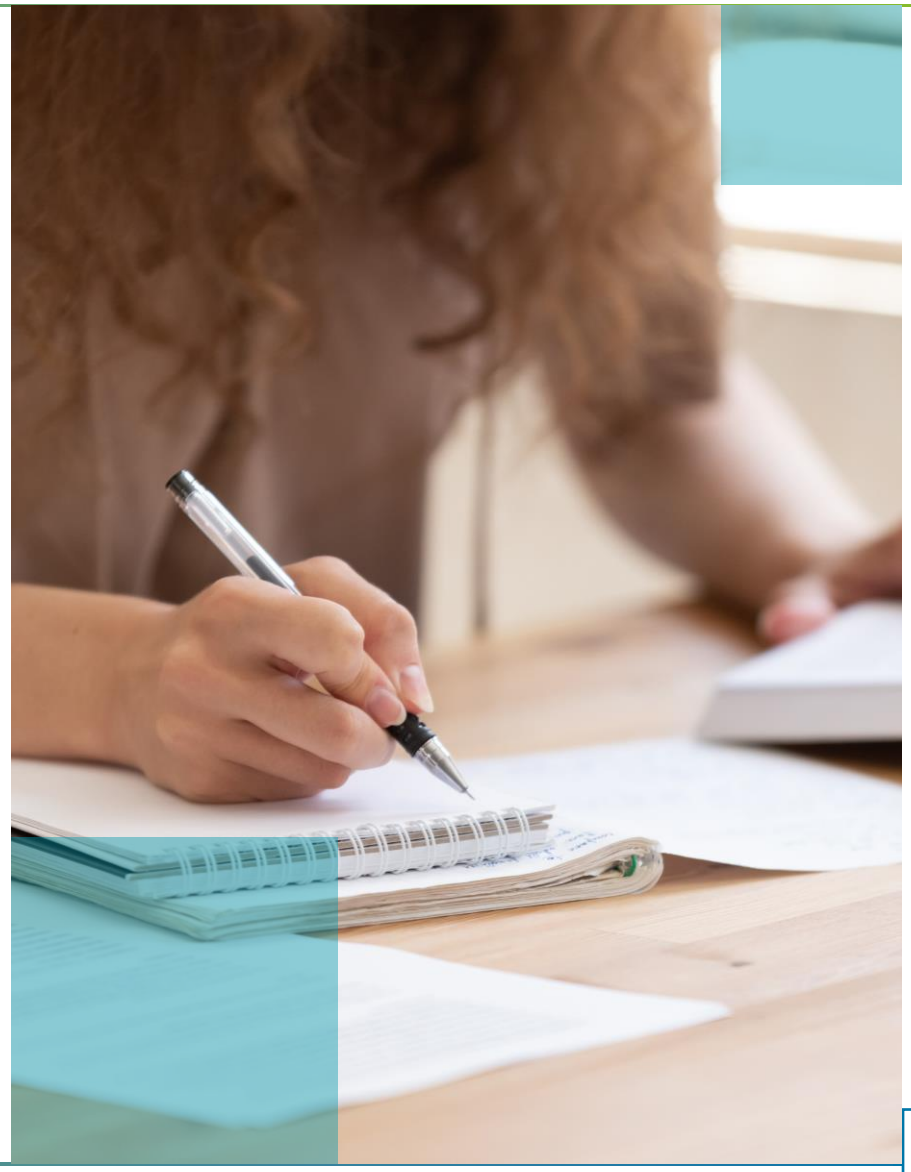
Recommended Preparation for a System Conversion Project

- Run the maintenance planner (https://apps.support.sap.com/sap/support/mp) to confirm that your SAP ERP system (including SAP and partner add-ons) can technically be converted to SAP S/4HANA.
- Explore in the simplification list (http://help.sap.com/S/4HANA_op_1610 → Simplification list) which of the functionalities relevant for you have changed. The simplification list is also available in Microsoft Excel Spreadsheet (XLS) format, see note 2313884.
- Start early to clean up your customer and vendor data in your current SAP ERP system in preparation for the business partner and Customer/Vendor Integration (CVI) in SAP S/4HANA. Further info in the simplification list.
- Check your custom code and optimize it for SAP S/4HANA. Further info on help.sap.com.



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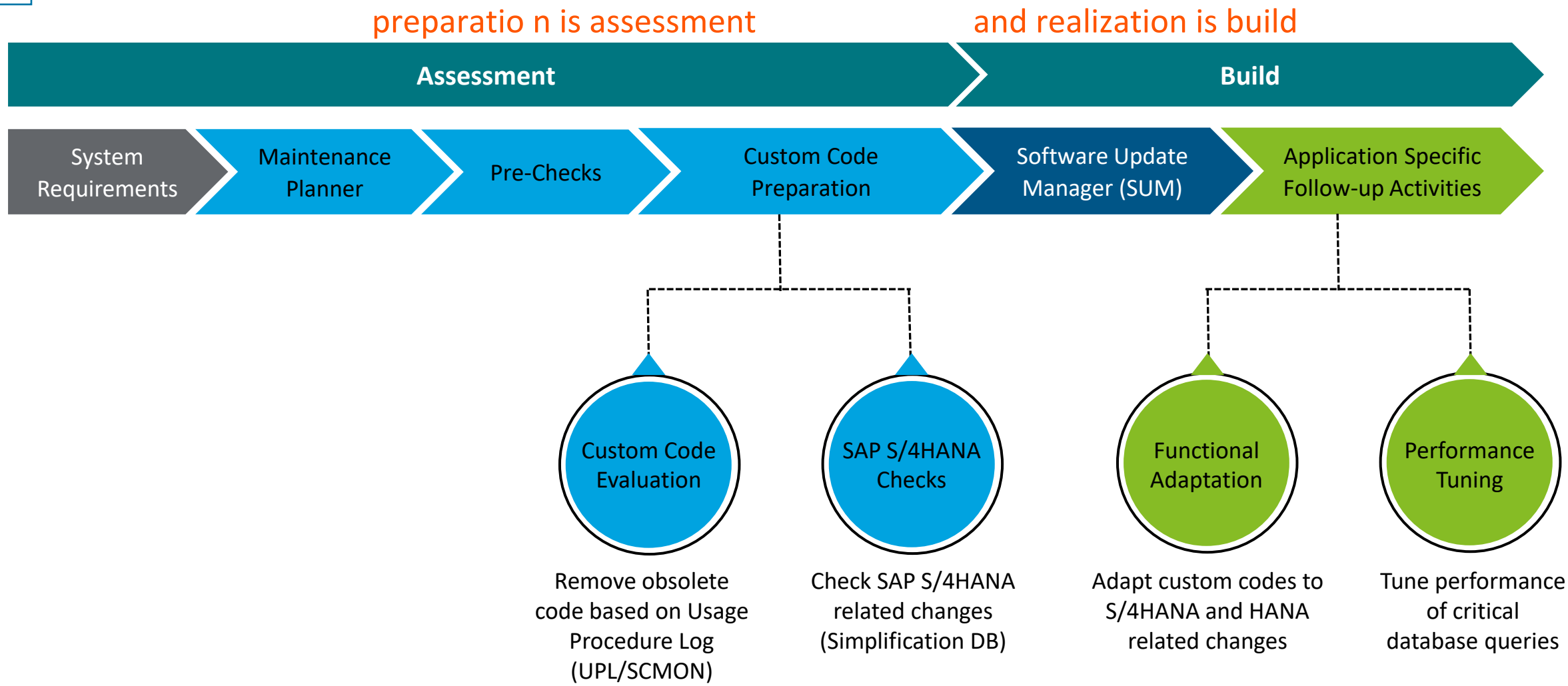
Custom Code Check and Application Specific Follow-up Activities



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Custom Code Checks

These checks are based on the Simplification list concept. Before converting to SAP S/4HANA check the custom code against the SAP S/4HANA simplifications. These simplifications are loaded into the custom code analyzer tool (transaction SYCM), where one gets the overview, searches for simplifications related to certain objects/object types, and navigates to the relevant SAP notes.

The tool of choice for SAP HANA and SAP S/4HANA checks is the Advanced Business Application Programming (ABAP) Test Cockpit (ATC) with Remote Code Analysis. After one runs the ATC, a list of findings is obtained where the custom code does not comply with the scope and data structure of SAP S/4HANA, on-premise edition.

Software Update Manager

Software Update Manager (SUM) is the technical tool for system conversion to SAP S/4HANA. Use the latest version of SUM 2.0 and run it after the maintenance planner.

Three steps are executed within SUM in one step:

Software Update	Update system with S/4HANA software
Database Migration	If it is required to migrate the database to SAP HANA database—the so called Database Migration Option (DMO) of SUM.
Data Conversion	Conversion of data from the old data structure to new S/4HANA data structure

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Functional Adaptation

- After system conversion to SAP S/4HANA is performed with Software Update Manager (SUM) (One need not first migrate the database to SAP HANA and then update the software to the SAP S/4HANA, the SUM does it in one step) it is recommended to run ATC with SAP HANA and SAP S/4HANA checks.
- After that one need to carry out functional adaptation based on the ATC results.

1 First one need to adapt the modifications and enhancements using the standard transactions `SPDD`, `SPAU` and `SPAU_ENH`.

2 Second, one need to fix SAP HANA and SAP S/4HANA findings.

- Adapt custom ABAP code, using the findings of the SAP HANA and SAP S/4HANA checks (ATC results). Findings of SAP S/4HANA checks are related to S/4HANA simplifications.
- Each simplification requires a different approach to solving findings. Therefore, findings of SAP S/4HANA checks refer to an SAP note which describes how to solve them.

Example: `ORDER BY, MATNR` (increased to 40 lengths)

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Performance Tuning

- After one does system conversion to SAP S/4HANA and the system is up and running, one needs to look at which business processes need to be optimized on the SAP HANA database since one now can make use of the full power of SAP HANA regarding performance. Therefore, one needs to look at which Structured Query Language (SQL) statements can be optimized.
- The SQL monitor (ABAP SQL Monitor–Implementation Guide and Best Practices) allows getting performance data for all SQL statements executed in your productive system. One can run it for a longer time period directly in a productive system (transaction SQLM) without a major performance impact on your business processes. (performance overhead < 3%).
- SQL monitor helps to understand, what are the most expensive and most frequently executed SQL statements, which SQLs read/write millions of records and provides the transparent SQL profile.
- SQL monitor allows to the link of the monitored SQL statements to the corresponding business processes including various aggregation and drill-down options.

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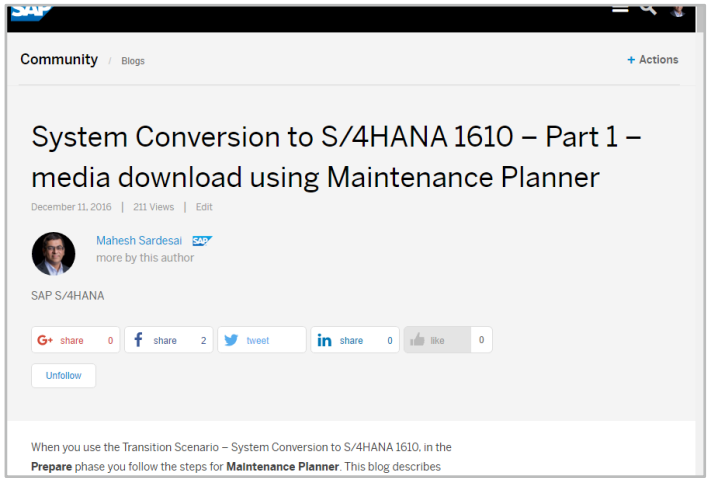
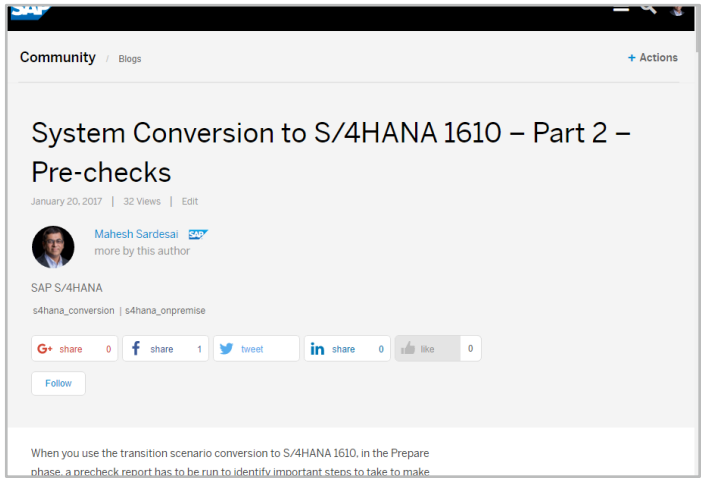
System Conversion

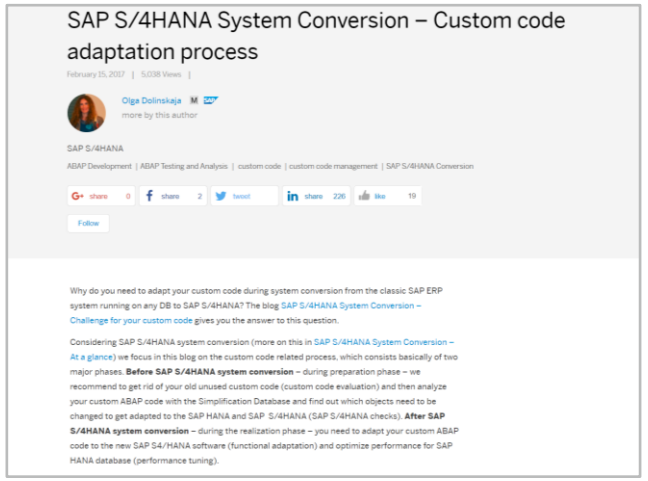
Example for SAP S/4HANA 1610



Maintenance Planner Blog



Pre-checks Blog



Custom Code Migration Worklist

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SAP S/4HANA Migration—Brownfield Approach An examples

Our S/4 migration global approach to this project was based on a six-stage methodology, that relies on five different system conversion cycles.

0 Assessment /POC

- Check feasibility for S/4 migration
- Migrate a copy of the production system to S/4HANA and test critical processes and impacted areas per the simplification list
- Identify a list of custom codes to be modified.
- Size/price new hardware

1 Prep

- Project roadmap
- Staffing plan
- Kick-off

2 Build

- Adjust existing custom code for S/4HANA and enable mandatory new functions as well as database changes
- Conduct unit tests
- Adjust security-related changes

3 Testing

Test the existing solution in converted S/4HANA system and obtain user sign-off

4 Mock Cutover

- Identify critical reports to compare pre-post conversion of system
- Dress and rehearse production cutover to understand the required outage and necessary steps

5 Go-Live

Migrate production system to S/4HANA, run critical reports (pre and post migration), and reconcile for migration sign-off

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Major Phases in System Conversion to SAP S/4HANA

Analysis

- Analysis of landscape requirements (e.g., used add-ons, third party integrations etc.)
- Review guides (simplification/conversion)
- Comparison of the existing solution with the S/4HANA product scope
- Evaluation of custom code against S/4HANA product scope
- Decision to be made if the system is eligible for system conversion or migration

Preparation

- Analysis of landscape requirements
- Assess system readiness (sizing etc.)
- Apply pre-requisite notes
- Perform functional preparation activities (including pre-transition reports)
- Evaluate custom code against S/4HANA product scope
- Build a detailed project plan, resource scheduling, cut over the plan, and post upgrade support plan

System Conversion

- Software updates (incl. S/4HANA application core installation)
- Execution of SUM—data model conversion
- SPAU/SPDD adjustments
- Mandatory configuration and custom code adjustments
- Upgrade all other systems (PO/Gateway etc.)
- Re-establish connections between S4 and all other systems
- Installation and implementation of Fiori (optional)

Post System Conversion Monitoring/Stabilization

- Monitor system performance
- Monitor all critical interfaces like banks/third party/ payroll etc.
- Single sign-on (If required)
- Monitor critical business processes

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High Level Assessment—Dev

Key Considerations:

Programs

Forms

Fiori/Gateway

Integrations

SPAU/SPDD Adjustments

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High Level Assessment—Dev (Cont.)

Programs

Simplification List	This is important activity to be performed to confirm whether custom code is correctly using S4 relevant.

Refer to the following SAP notes:

- **2241080**: Instructions to download the latest support package of the simplification database from the SAP marketplace
- **2185390**: Custom code analyzer
- **RS_ABAP_SOURCE_SCAN** is used to generate detailed source code analysis of objects highlighted in the simplification list.

RS_ABAP_SOURCE_SCAN is a se38 program which searches for a given string in all programs

Additional notes referred for simplification list analysis:

2198647	S/4HANA data model changes in SD
2206980	Mat. inventory manag. change of data model in S4
2215424	Mat no. field length extension-general information
2215852	Mat. no. field length extension code adaptions
2217299	Inv. valuation-inv management change of data model in S4
2223144	S/4HANA foreign trade in SD_MM

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High Level Assessment—Dev (Cont.)

Programs (Cont.)

- Perform syntax checks with Sensitive Compartmented Information (SCI) (SAP code inspector) and ABAP Test Cockpit (ATC). Use the following variants:

```
FUNCTIONAL_DB
```

```
FUNCTIONAL_DB_ADDITION
```

```
PERFORMANCE_DB
```

- Need to perform TUT/FUT for objects which are changed

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High Level Assessment—Dev (Cont.)

Forms

SAP script and smart forms are still supported in S4. The preferred technology for forms is still Adobe forms

FIORI/Gateway

- Old Fiori apps are replaced by new S4 Fiori apps. The developer needs to review the S4 Fiori apps library in detail and review it against existing Fiori apps. For e.g., Purchase order/purchase req approval apps are replaced by the My Inbox app in S4.
- Fiori developer needs to work in collaboration with basis and security resources to configure and test all in scope Fiori apps.

Integrations

- Test all interfaces/integration points to confirm SAP PO 7.5 impacts
- Conduct end to end testing with a few third parties to confirm various adapters

SPAU/SPDD Adjustments

- Performing SPAU/SPDD adjustments is an important step in the conversion process
- Depending on the business functionally required, the developer has to assess and perform actions
- Modify custom objects as per instructions
- Need to perform TUT/FUT for objects which are changed

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Advantage of Brownfield System Conversion

System Conversion: It is a complete technical in-place conversion of an existing SAP business suite ERP system to SAP S/4HANA. This is the preferred option for customers who want to bring their business processes to the new platform and want to adopt new innovations at their speed, then.

A Brownfield conversion presents a risk-free opportunity to cost-effectively combine the SAP landscape rebuild with an overall consolidation and harmonization of all SAP-related business processes.

- If you are already using an SAP solution, you may be completely happy with its performance and stability of it. You may have suffered many issues during your SAP journey and finally, things have settled down and users are happy. So now, you are being told to upgrade or migrate to S/4HANA.
- System conversion, also known as the ‘Brownfield’ approach, enables migration to S/4HANA without re-implementation and without disruption to existing business processes. At the same time, it enables the reevaluation of customization and existing process flows.
- "Most organizations might seek a way to safeguard the investments they have made over time to continuously improve and configure their SAP environment.
- The SAP S/4HANA system conversion is the scenario to be chosen if a customer wants to change their current SAP ERP system into an SAP S/4HANA system. Based on the offered tooling technical the SAP S/4HANA software is installed, the database is migrated to the SAP HANA database (if not already done), and transfers data from the old data structure into the new S/4HANA data structure (e.g., migrate data into the new MATDOC table) in one-step.

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SAP S/4HANA—Transition

Scenario 3—Landscape Transformation (1/2)

Scenario Description: Customers who want to consolidate their landscape or selectively transform data into an existing SAP S/4HANA system.

Why Would You Choose This option?

Transactional Version Memory	Selective data transformation allows a phased approach focusing the first SAP S/4HANA migration phase on parts of the business with the highest Return On Investment (ROI) and lowest Total Cost of Investment (TCI)
Agility	Stay on current business processes but move gradually to SAP S/4HANA innovations ⟶ Move to SAP S/4HANA at your own pace
TCO Reduction	System and landscape consolidation with harmonized/simplified processes and unified master data lead to lower cost of operations

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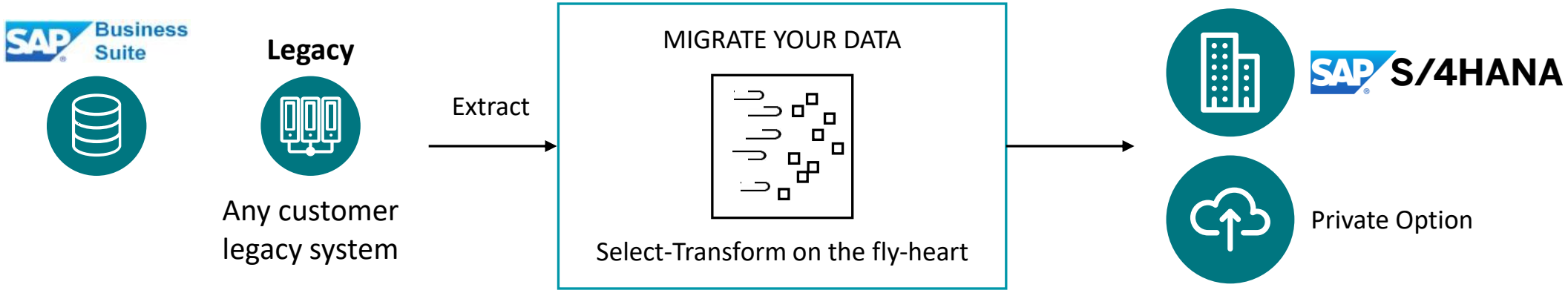
SAP S/4HANA—Transition (Cont.)



Technical migration on table level using pre-configured LT transformation solutions

Consolidation	System merge of multiple source systems into one SAP S/4HANA system or transfer of clients into existing SAP S/4HANA system (build-up multiple client system)
Selective Data Transformation	Migration of business units/single entities such as company code Re-platforming scenarios, e.g., PP/DS

Real-time Reposting:

Implementation of central finance: Real-time reposting of financial transactions into central finance instance, e.g., to enable centralized reporting using SAP S/4HANA (synchronization of systems/side-car scenario)

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SAP S/4HANA—Transition (Cont.)

Scenario 3 – Landscape Transformation (2/2)

The pre-configured landscape transformation Scenarios can migrate data

- into newly implemented SAP S/4HANA target systems
- as well as into an SAP S/4HANA system created via a system conversion procedure
- before or after they are handed over to operations
- and independent if they are foreseen for productive use or not

The pre-configured landscape transformation scenarios can complement the other SAP S/4HANA transition options.

Note 1: SAP landscape transformation scenarios are supported within SAP S/4HANA value assurance engagements. The transformation solutions mentioned for technical transformation are not generally released.

Note 2: SAP is ready to discuss and support data migration requirements beyond the pre-configured use cases. Refer to the data migration services in the activate phases prepare and explore for details.

Transition to S/4HANA Roadmap Viewer

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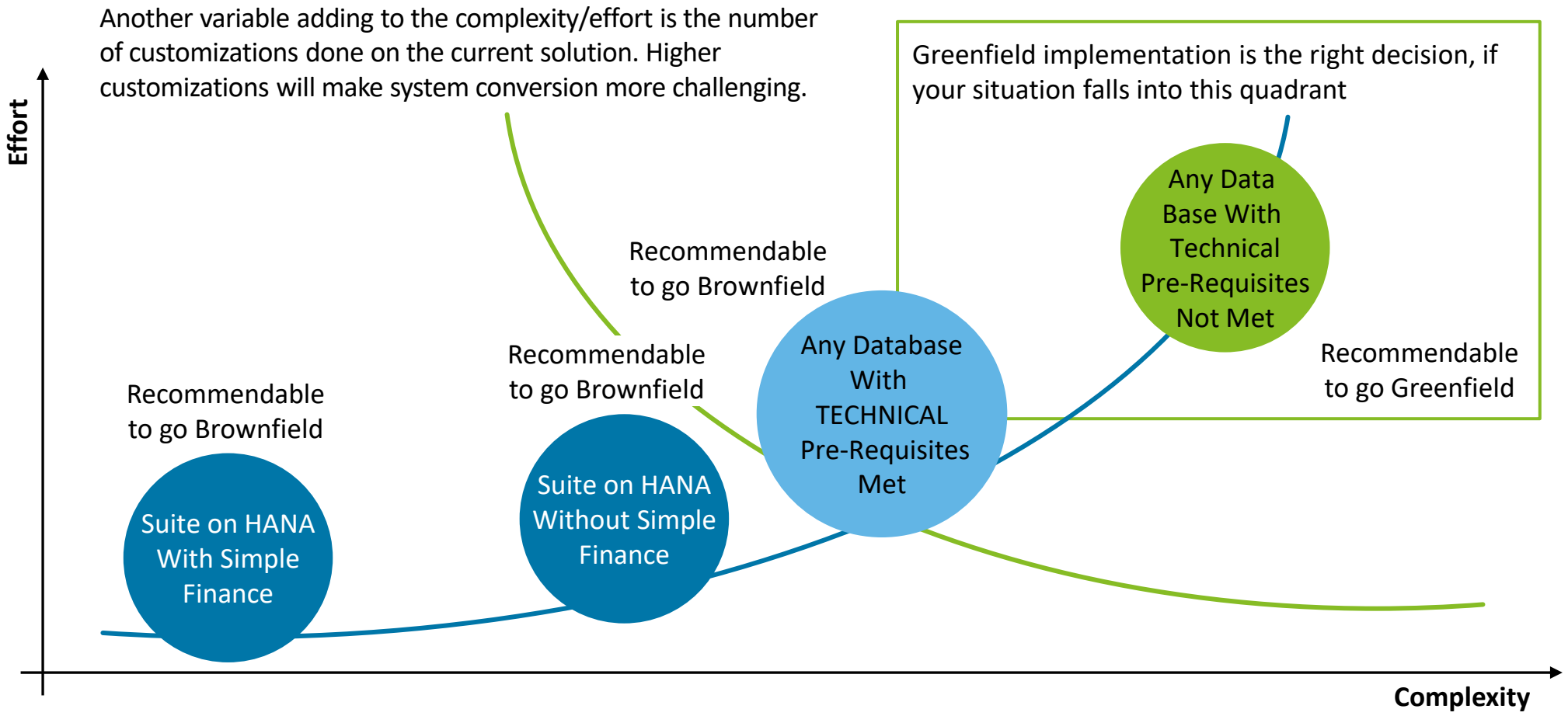
Influencing Factors—Brownfield or Greenfield?

Below we have our graphical depiction of effort vs complexity with various versions of SAP for system conversion or greenfield approach to S/4HANA decision making.



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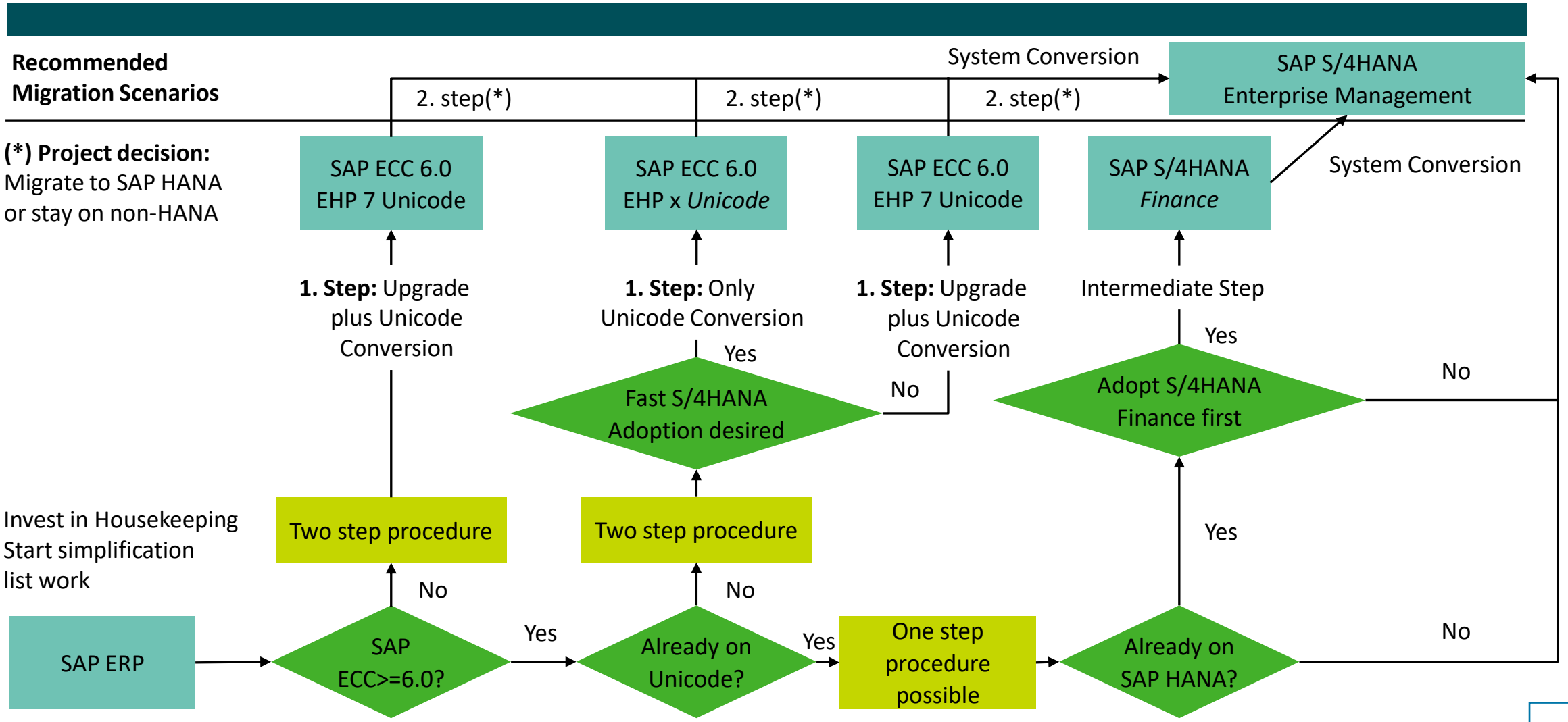
Recommended Migration Scenarios



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Roadmap Options for S/4HANA

When it comes to preparing our companies to transition into S/4HANA, we have two different transformation scenarios available. So, if you are planning to migrate one or more SAP systems to S/4HANA, you have to think about whether it is going to be a brownfield or a greenfield, weighing up the pros and cons of both approaches.

System Conversion (Brownfield)

- The current template is taken as the basis for the new implementation
- In the S/4HANA context, brownfield classically implies a “big bang” in one instance
- Plan quick design sessions for new best practices and pre-requisites (e.g., Business Partners)
- Big bang to be done on the full landscape with a big testing effort. All countries would run on S/4HANA

Example

Complete conversion of an existing SAP business suite software system to SAP S/4HANA

The **key advantage** is generally a shorter runtime and less disruption of operational activities because you keep closer to your current way of working

New Implementation (Greenfield)

- A new SAP System is implemented based on a fresh installation of the SAP S/4HANA. Full innovation as part of the new build
- One or a set of Business Units (Bus) can be on S/4 instance first and then the rest can move in
- Rollout sequence and clustering fairly flexible
- Need to maintain both environments for the duration of deployment to S/4HANA

Example

Existing SAP customers implementing a new SAP S/4HANA software system

The **key advantage** is generally that you start with a new system and therefore have more flexibility to introduce themes such as standardization and simplification in addition to the migration itself

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Greenfield vs. Brownfield Migration Analysis

		Brownfield	Greenfield
Business Benefits	• Leverage existing process and extend the benefit on S/4HANA Core • Reimagine new simplified business processes in SAP S/4HANA	^	^
Technical Implication	• Existing business functions are compatible to SAP S/4HANA • Opportunity to establish “Clean ERP” with “Platform First” approach	^	^
Change Management	• Existing processes, code, and screens can be leveraged with retrofits • The processes that runs on Fiori based apps is the key to user adoption	^	^
Tech Adoption	• Carry over existing code base by adjusting the conflicts w.r.t simplified DB model • Build S/4HANA compatible code and replace customizations with standard functionality/Fiori	^	^
Time Line And Upgrades	• Recognize immediate benefit of S/4HANA with faster pace of implementations • Adopt standard functionality to enable seamless and faster upgrade to next S/4HANA versions	^	^

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Data Consideration for SAP S/4HANA Transition

- SAP data between R/3 or ERP Central Component (ECC) will be different in S/4HANA.
- In some areas there is consolidation (business partner) while in some areas its new functionality (credit data going into Financial Supply Chain Management (FSCM)).
- Clean data is needed to consolidate as well as migrate to new areas.
- Most clients with legacy SAP systems have built up data that must be cleansed or archived before migration to S/4HANA.

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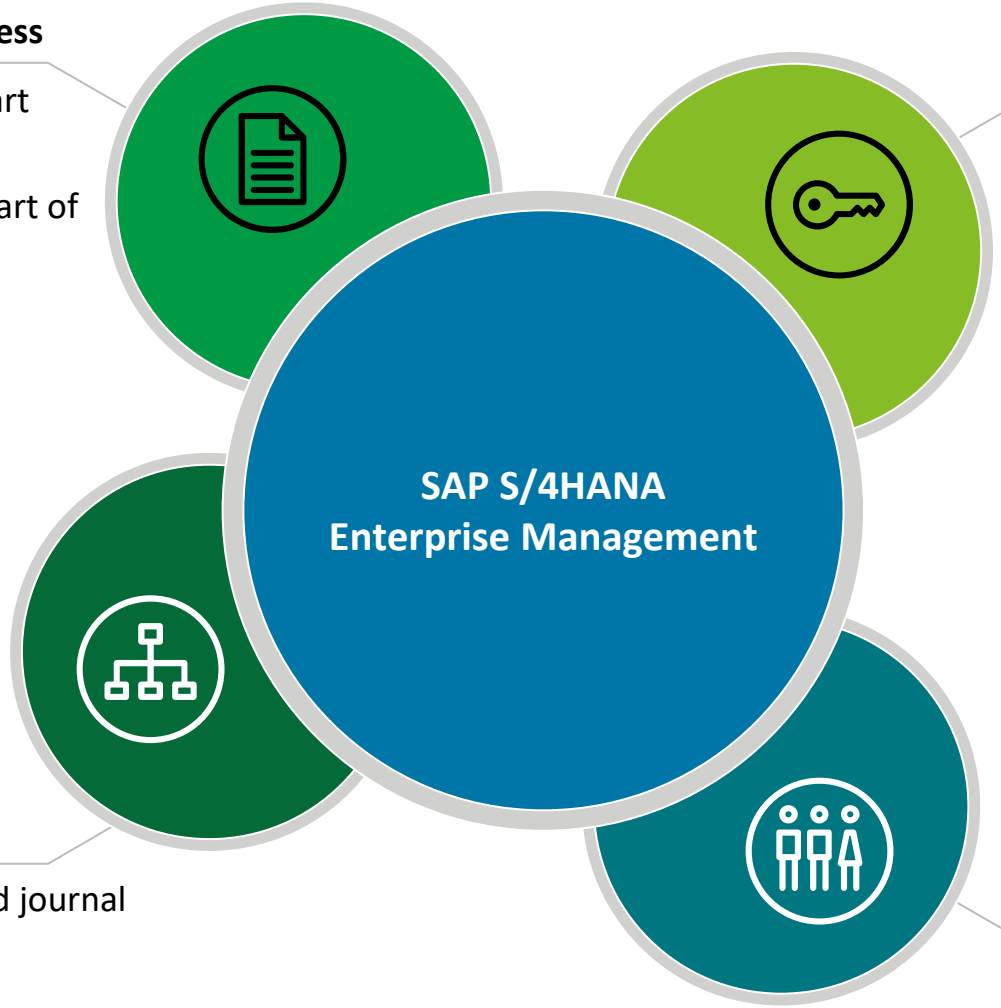
Data Consideration for SAP S/4HANA Transition (Cont.)

Realigned applications to Business

- Credit Management is now part of FSCM
- Revenue recognition is now part of revenue accounting

Other Key Changes

- Foreign trade is replaced by Global Trade Services (GTS)
- Mat. Ledger is obligatory. Costs may now be maintained in multiple currencies for a material



Simplified Data Models

- Accounting data with a unified journal
- Material movements
- Sales and distribution documents

Mandatory Centralized Partner Management

Customer, vendor, employee, credit, etc.

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Summary

Here are the key learning points of the module.

- SAP S/4HANA is the digital core for IT transformation.
- Aggregation is now executed dynamically in real-time by the HANA database and reverse compatibility is enabled through HANA compatibility views.
- SAP HANA architecture is a flexible, data source-agnostic in-memory appliance to analyze large volumes of data in real time.
- S/4HANA enterprise management introduces a new accelerator product titled SAP activate that looks to reduce the time and effort needed to implement S/4HANA enterprise management.
- The SAP S/4HANA migration object modeler is part of the SAP S/4HANA Migration cockpit.
- Maintenance planner simplifies the landscape maintenance process to update, upgrade, and install systems with ease and efficiency by checking the system with regard to business functions, industry solutions, and add-ons.
- A Brownfield conversion presents a risk-free opportunity to cost-effectively combine the SAP landscape rebuild with an overall consolidation and harmonization of all SAP-related business processes.

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